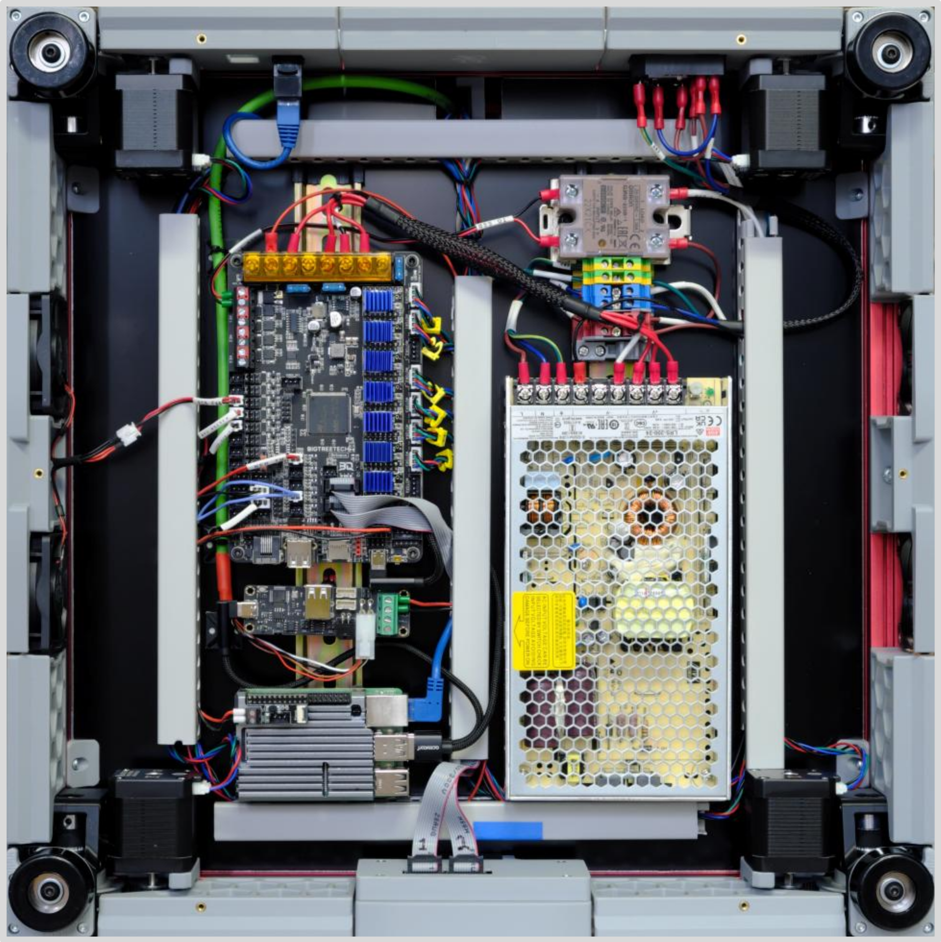
An Introduction to 3D Printing

Part 1: FFF Printers

Introduction, Process, Limitations, and Advantages

About Me

- Learned 3D printing by building my own printer.
- Voron 2.4 GitHub.
- Fully open-source, DIY project.
- 6k hours (250 days) of print time.
- printables.com Flying Gyroscope



FAQ: Which Printer?

Which printer should I get?

- Get the most **reliable** printer for the amount of money you have.
  - Do not get sucked into marketing.
  - Do not obsess over spec sheets or speeds.
- Is the printer the hobby, or is making things the hobby?

FAQ: Which Printer?

- Do you want open-source?
  - Community projects. (Do not settle for poor documentation.)
  - Repairability.
  - Upgrades.
  - Leading edge innovations.
- Do you want curated “just works” experience?
  - Top brands have excellent slicer profiles (settings) – **highly** recommended for beginners.

FAQ

How do I tune my printer?

- Make one change and try A vs B blind test.
  - Can you actually see or measure a difference?
- Resist the temptation to play with slicer settings that already work.
  - Print with default settings and keep the printer unmodified.
  - Then it is easier to diagnose problems.
- Ellis' Print Tuning Guide.
- OrcaSlicer Calibration Walkthrough.

Outline

Part 1: Printers

Process

Limitations

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

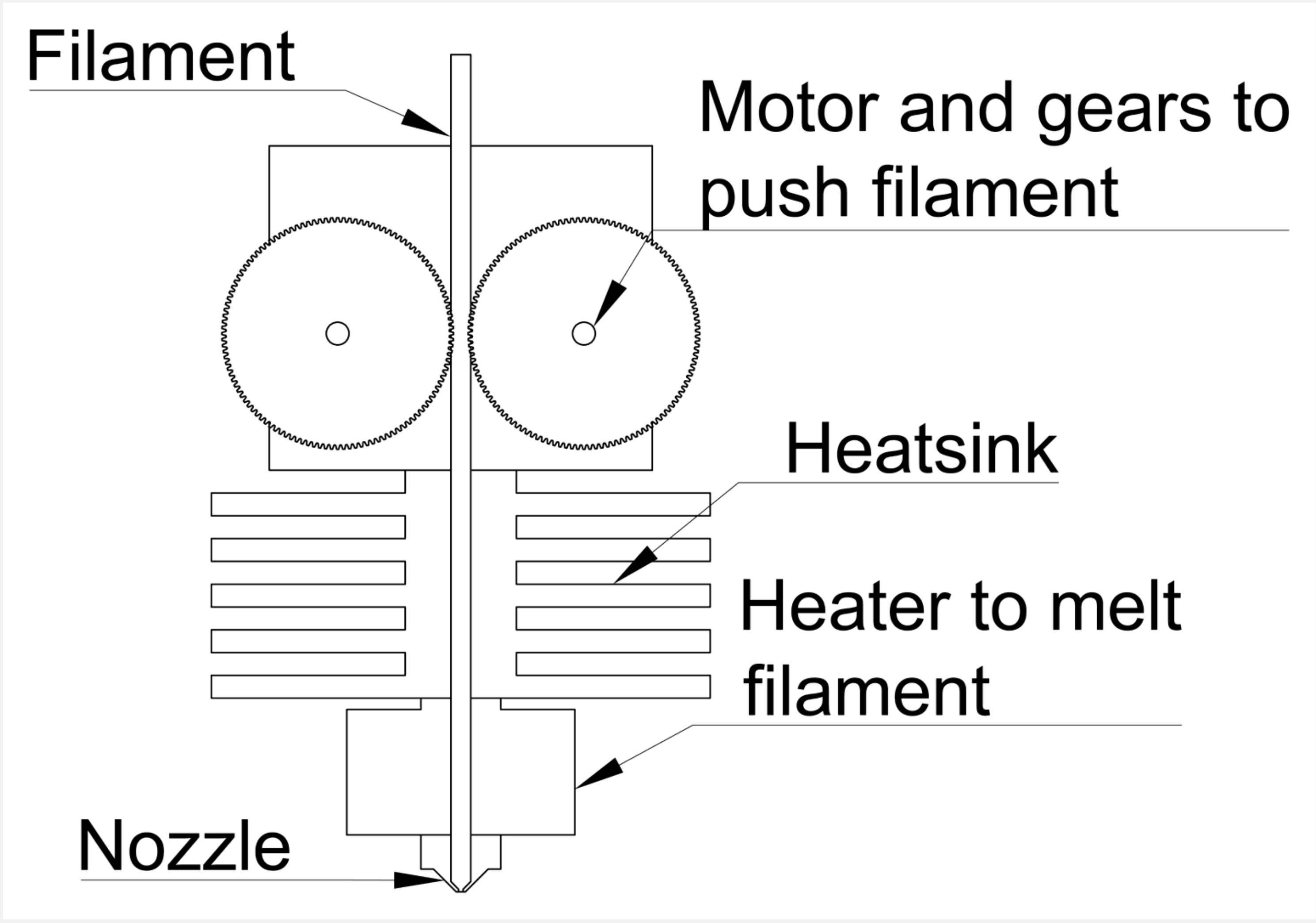
Thermal Expansion

3D Printing is a Process

1) Start with a 3D design (digital file).
   - If you download, check the license for restrictions.
2) Slice the design.
   - Printers do not accept 3D models.
   - Use software to “slice” objects into tiny layers.
   - Translate geometry into to gcode – *somewhat* standardized commands.
3) Print.
   - How did it turn out?
4) Post-process.
   - Remove supports, assemble, glue, sand (to smooth layer lines), paint, …

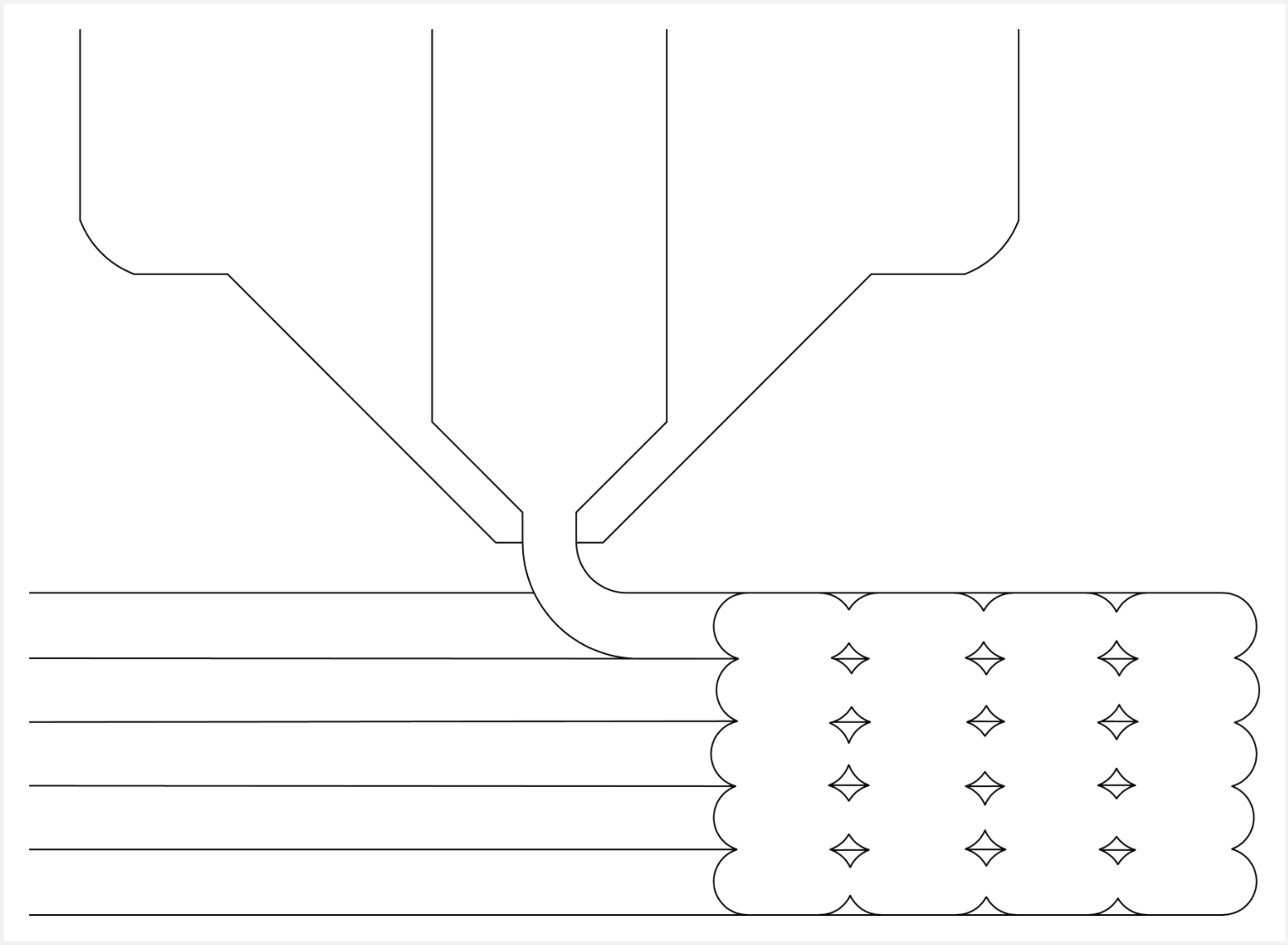
Pushing Plastic

- What kind of 3D printing?
- **FFF** (fused filament fabrication).
- Filament – long, continuous plastic “noodle.”
- Toolhead – the thing that moves around in X and Y.
- Cooling fans not shown.



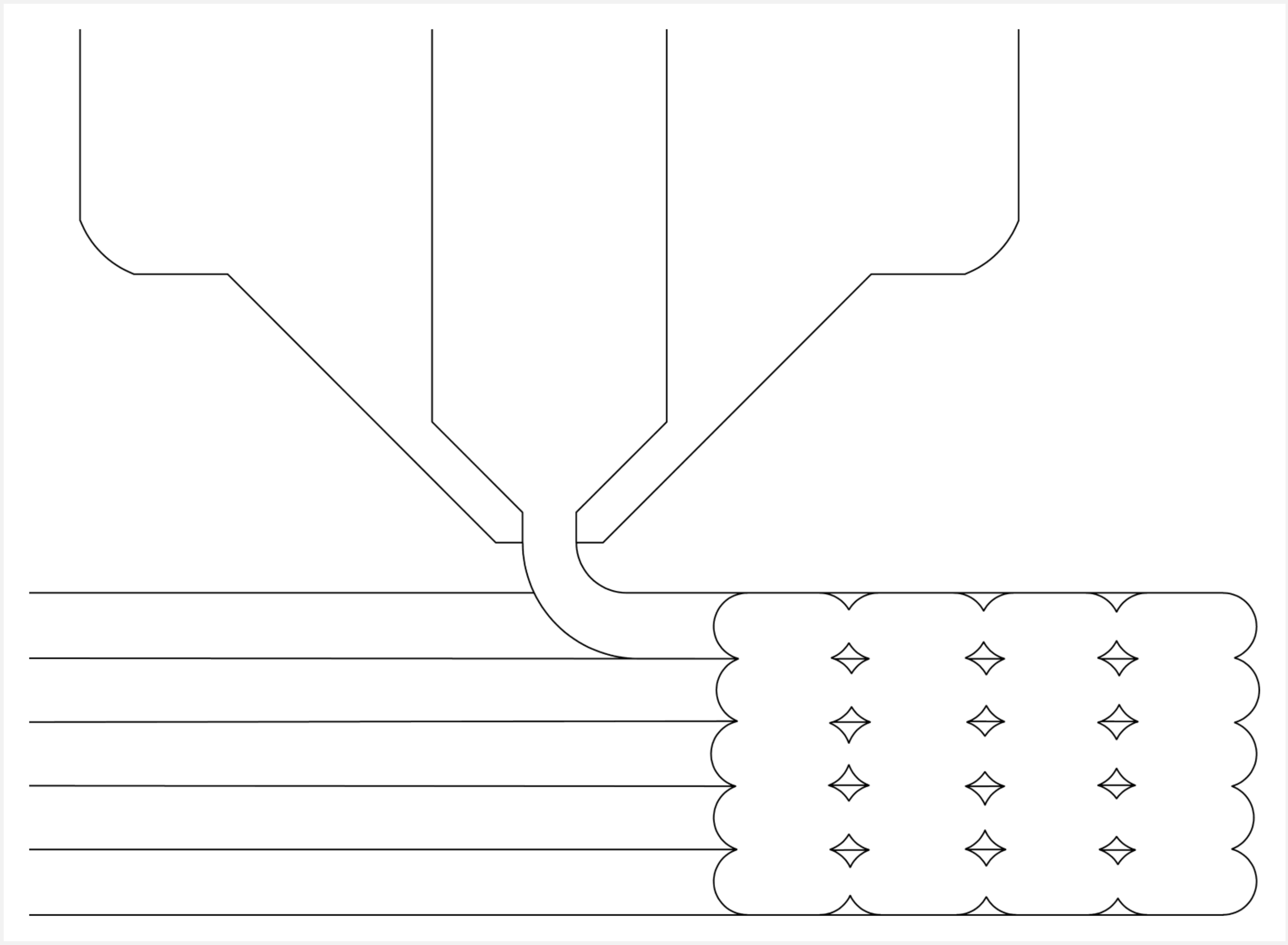
Extrusions

- The printer deposits a line called an **extrusion**.
- Squish extrusions into place (ideally).
- Extrusions lock in place when they cool.
- Need a fast cold-hot-cold thermal cycle.



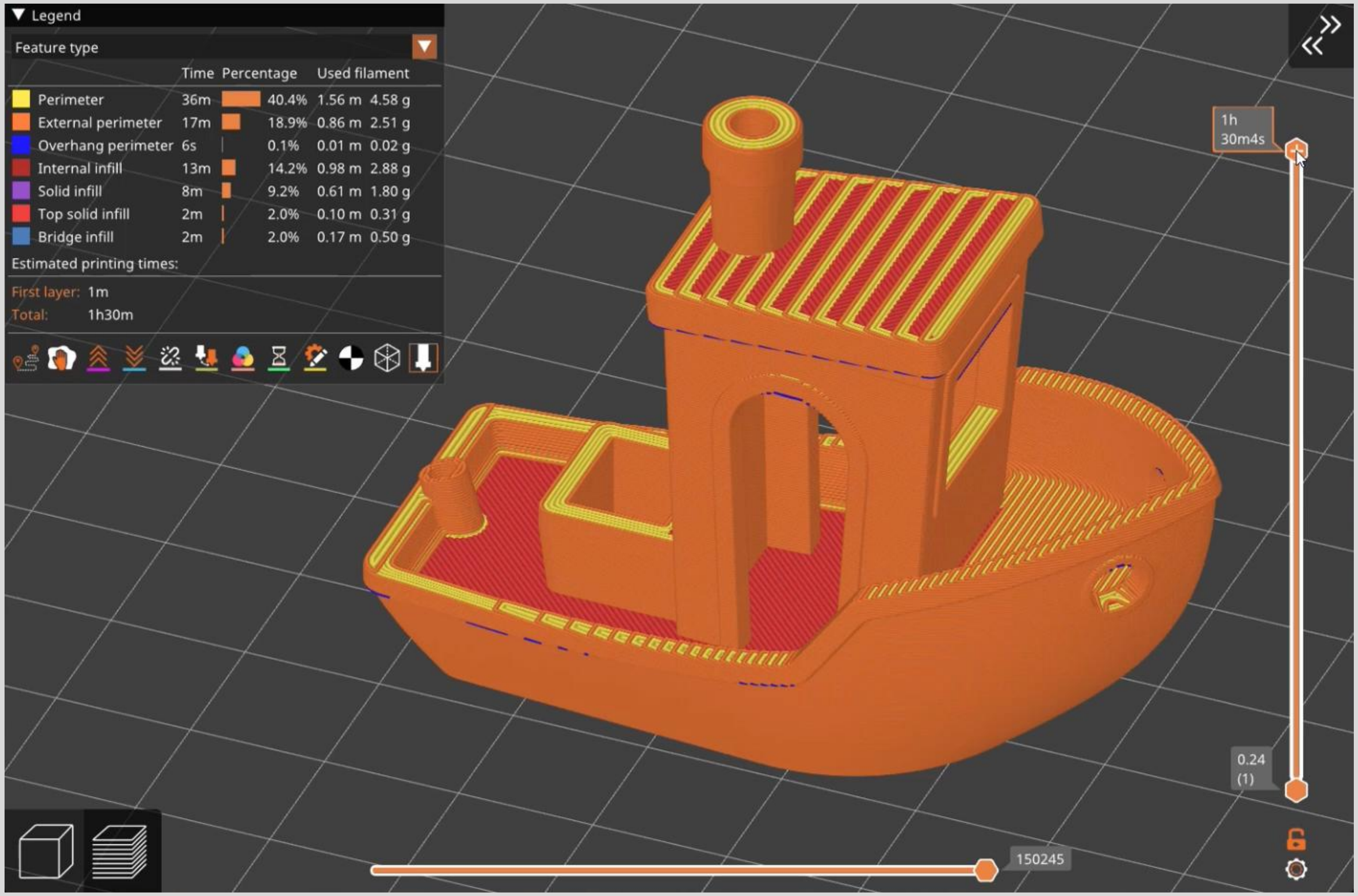
Stacking Layers

- Use extrusions to make layers.
- Stack layers to build up height.
- Printers *simulate* 3D shapes.
- You could call it 2.5D printing.



Slicer Layers

Let's see how PrusaSlicer makes layers.

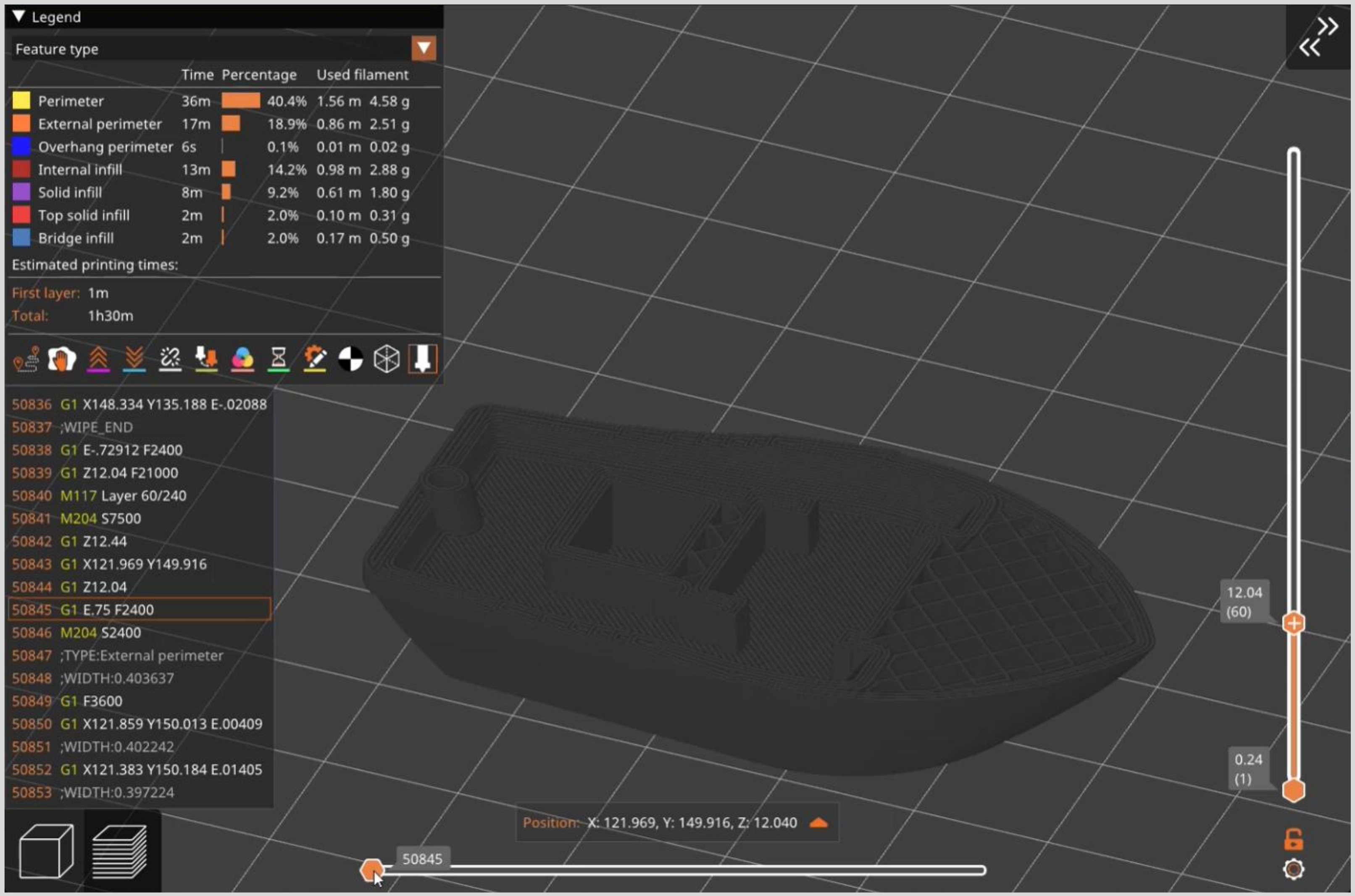


FlyingGyroscope GitHub.

Slicer Layers

Let's preview a single layer.

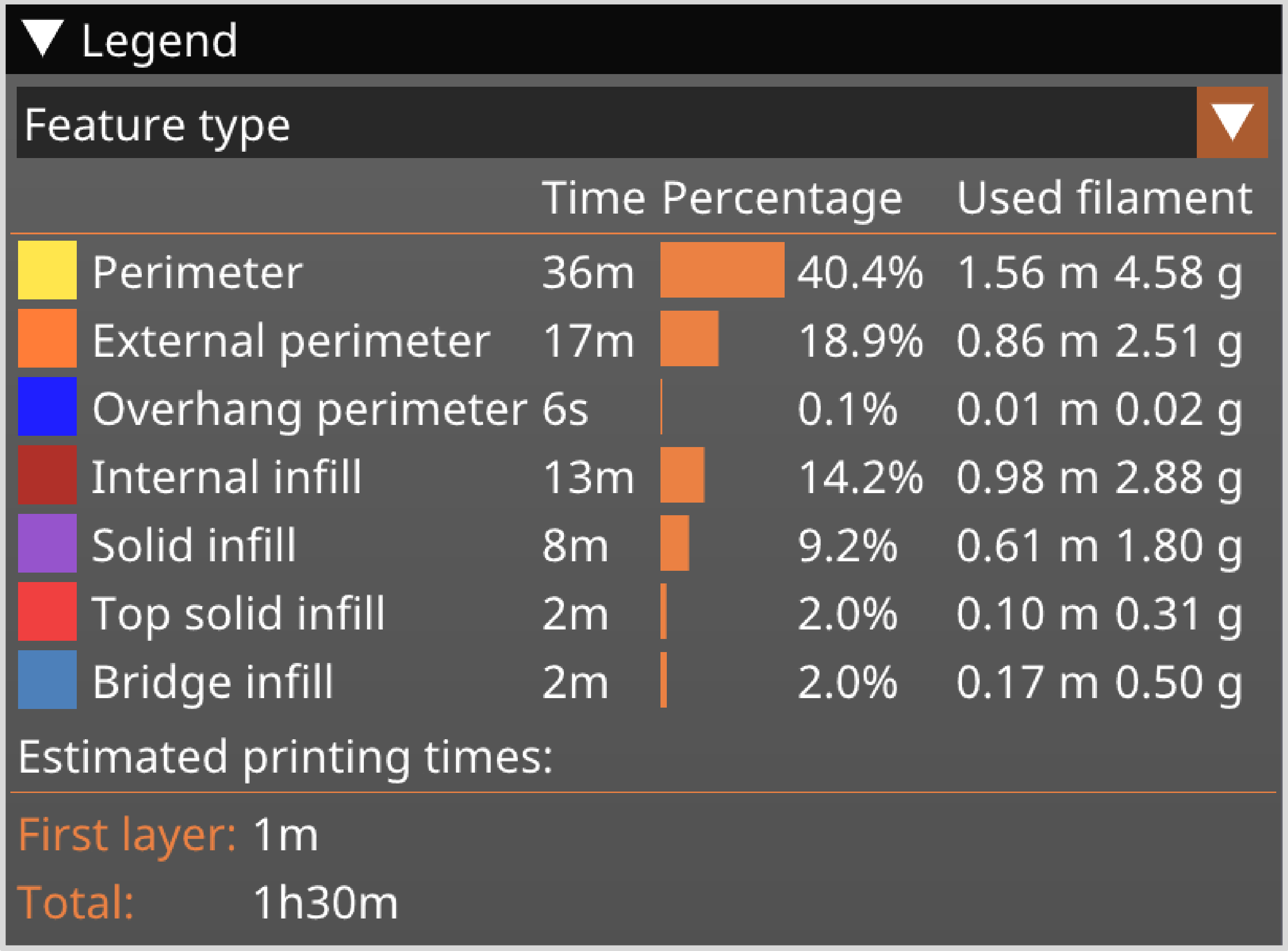
- Gcode preview (lower left) shows what the printer is actually doing.
- *G1* – move. Can include X, Y, or Z coordinates.
- *G1 E* – move and extrude (print).



[FlyingGyroscope GitHub]().

Slicer Layers

- Slicers sort extrusions according to geometry and shape.
- Apply optimized settings for each category. Typically …
    - Printing speed.
    - Extrusion width.
    - Cooling (fan speed).



Legend

Feature type

	Time	Percentage	Used filament
Perimeter	36m	40.4%	1.56 m 4.58 g
External perimeter	17m	18.9%	0.86 m 2.51 g
Overhang perimeter	6s	0.1%	0.01 m 0.02 g
Internal infill	13m	14.2%	0.98 m 2.88 g
Solid infill	8m	9.2%	0.61 m 1.80 g
Top solid infill	2m	2.0%	0.10 m 0.31 g
Bridge infill	2m	2.0%	0.17 m 0.50 g

Estimated printing times:

First layer: 1m

Total: 1h30m

(Thermal camera video by NeedItMakeIt.)

Thermal Cycle

- “Melting” filament is technically incorrect.
  - Why? There is no transition into a liquid (in 3D printing applications).
- Filament is **softened** into a viscous state, like a gel.
- “Glass transition temperature” instead of melting temperature.

Easy Filaments

- **PLA** (derived from corn starch).
  - General-purpose filament for indoor applications.
  - Very stiff.
  - Deforms permanently under load (creep).
  - Low temperature resistance — will not survive inside a car on a hot summer day.
- **PETG** (plastic water bottles).
  - Pretty good general-purpose filament.
  - Hygroscopic (needs low-humidity environment).
  - Excellent choice for springs (elastic).
  - Moderate temperature resistance — can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
  - Low warp (low shrink).
  - High dimensional accuracy.
  - Room temperature print chamber.

3D Printer Protocol

Let's review some simplified rules.

- Secure the loose end of a spool of filament **at all times**.
  - Tangles ruin prints and spools, and waste time
- Do not reach into the printer while it is running.
  - Use the touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Do not burn your hands.
  - Read temperatures on the display before you touch.
  - Do not reach near a hot nozzle (210 °C, 410 °F).
- Do not leave fingerprints on print sheets.

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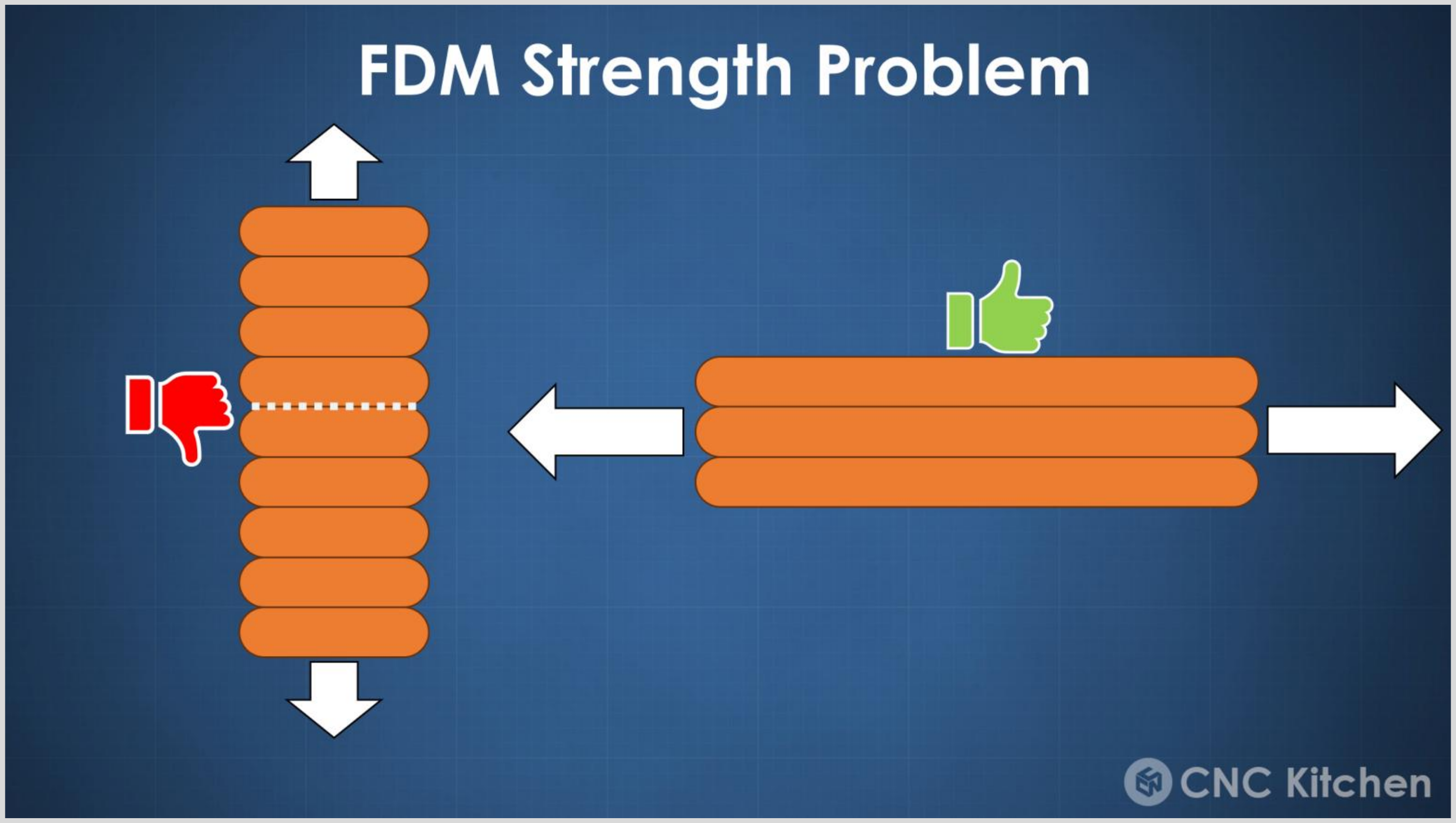
Thermal Cycle

- Some thermoplastics **shrink** as they cool, up to 1-2%.
  - Distorts and skews dimensions.
  - In extreme cases leads to cracking and delamination.
- Advice?
  - **Tightly control ambient temperature and cooling.**
  - Some filaments (ABS, Nylon) like it hot – extra heating.
  - Some filaments (PLA) like it cold – extra cooling.
  - PETG is somewhere in-between.

Anisotropic Strength

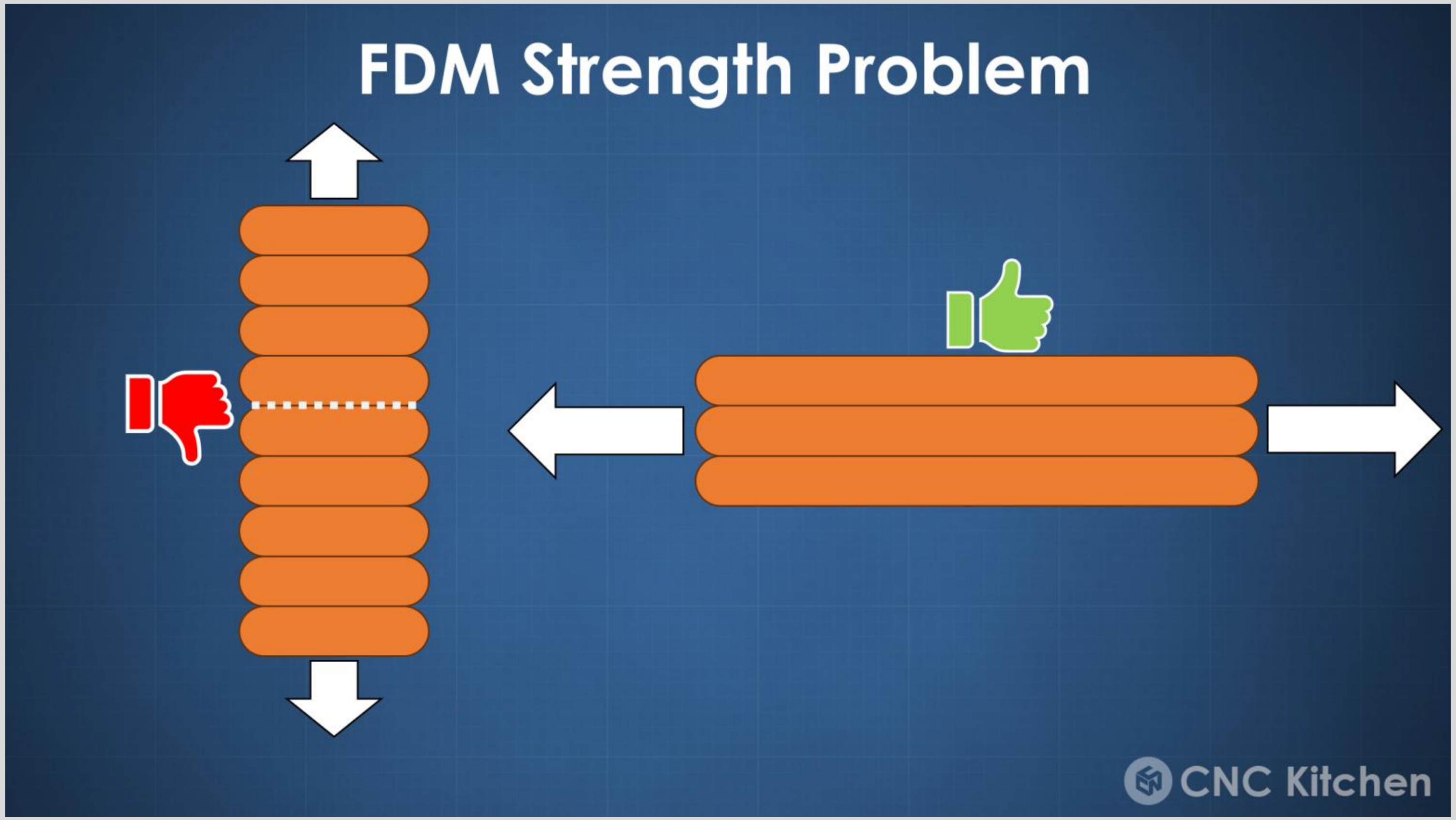
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament bonds to a cold/warm layer.



(Picture from CNC Kitchen.)

Anisotropic Strength

- Advice?
- **Try to avoid tall and skinny shapes.**
- **Lose around 50% of strength.**



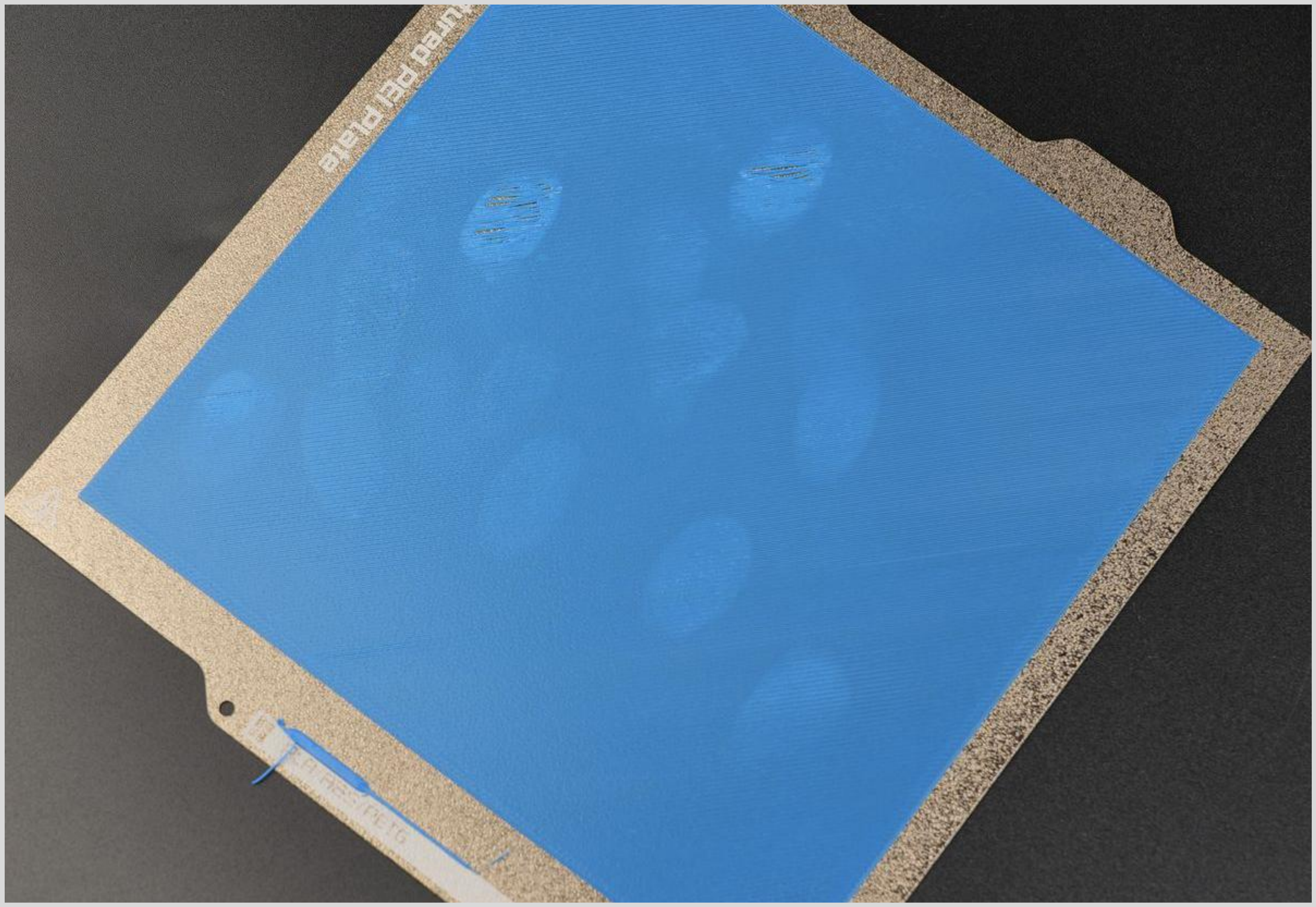
(Picture from CNC Kitchen.)

First Layer

- What is the most important layer in a 3D print?
  - The first layer!
  - First layers keep everything locked in place.
- Advice?
  - **Always watch the first layer.**
  - Tune first layer squish.
  - Flat surface area on the bottom of the 3D object.
  - Heated print bed.
  - Perfectly clean print bed.
  - Apply adhesion promoters (glue).

First Layer Mistakes

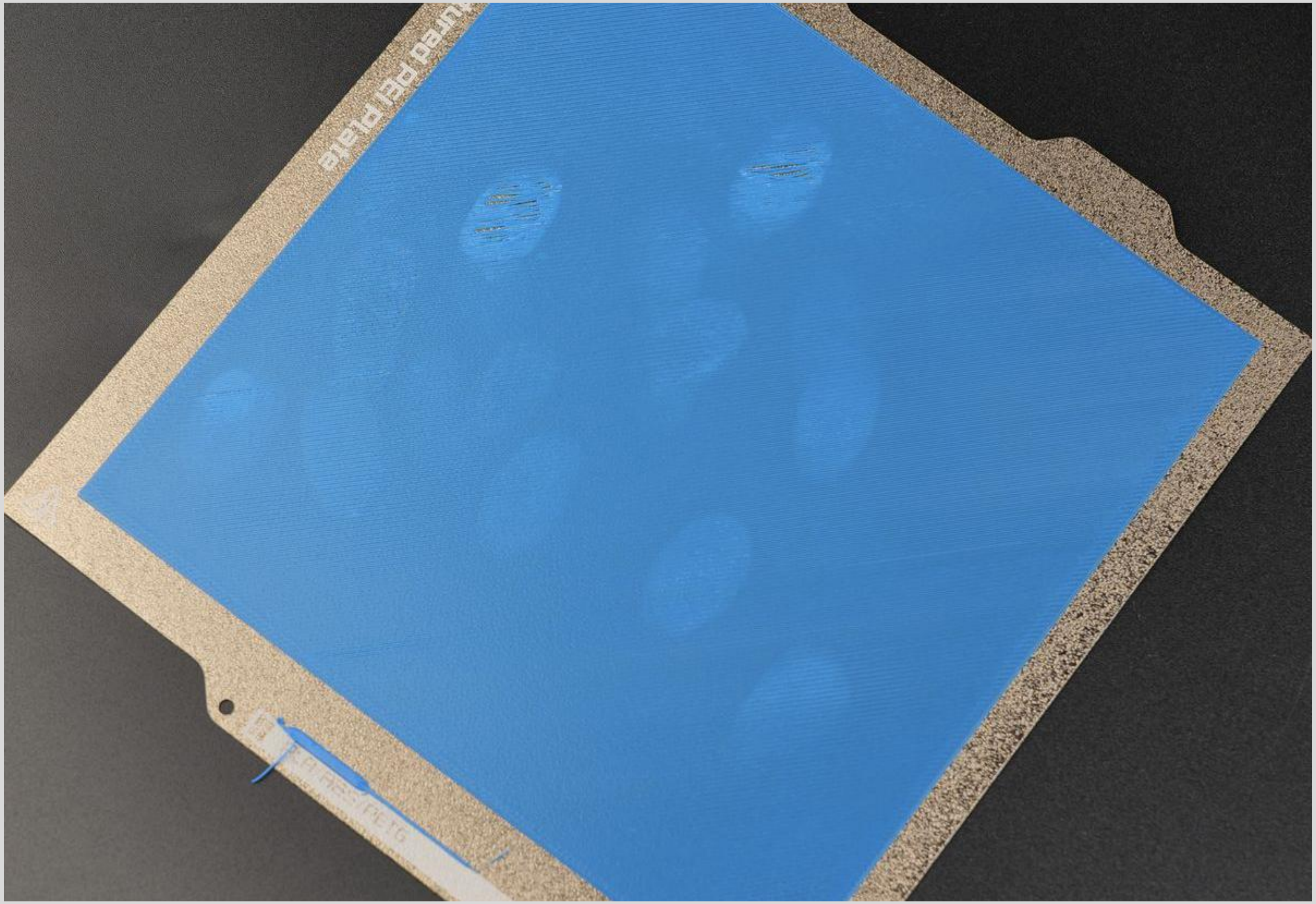
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from Bambu Labs Wiki.)

First Layer Mistakes

- Advice?
- Deep cleaning: **dish soap, brush, and water.**
- Everyday use: **window cleaner with ammonia.**



(Picture from Bambu Labs Wiki.)

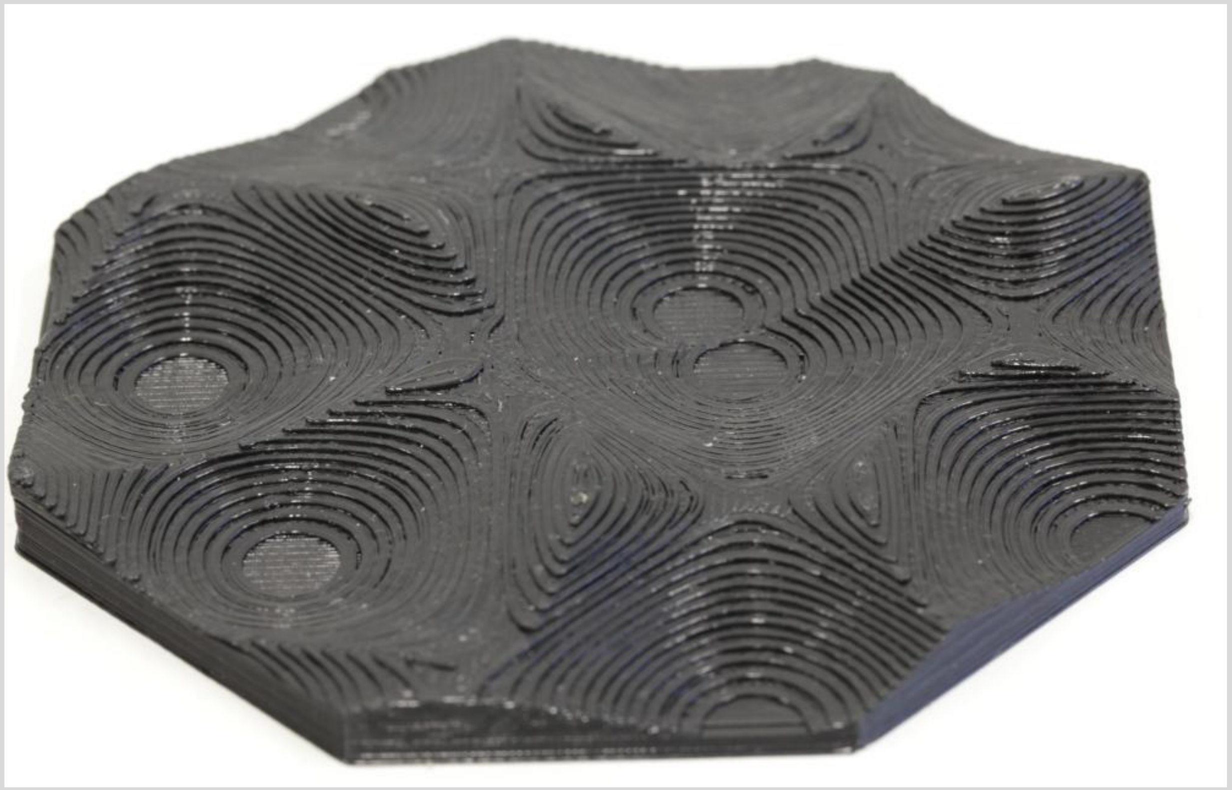
First Layer Mistakes

- What happens when a first layer does not stick?
- The print breaks free and makes "spaghetti".
- You can see evidence of when things went wrong – some of the rectangular print finished.

(Picture from Everything STEM.)

Stair-Stepping

- Printing with layers → Height changes in discrete steps.
- Mostly cosmetic effect.
- Advice?
- **Rotate objects to hide layer lines.**
- **Smaller layer heights — reduce visibility.**
- **Variable layer height for curved surfaces — gives a more consistent appearance.**



(Picture from Hamburg University Research Project.)

First Layer Warping

- Why do corners and edges peel up?
  - Temperature gradient from top to bottom.
  - Shrinking at different rates.
  - Internal stresses.
  - Stresses concentrate in corners.
- Shrinkage is a percent.
  - Large parts have the most trouble.

(Picture from Creality Cloud Blog.)

First Layer Warping

Consider an analogy.

- Imagine you want to remove a label or piece of tape. What do you do?
- You start peeling up the corner because that is the weak point.
- Same thing here – Physics found the weak point.

(Picture from Creality Cloud Blog.)

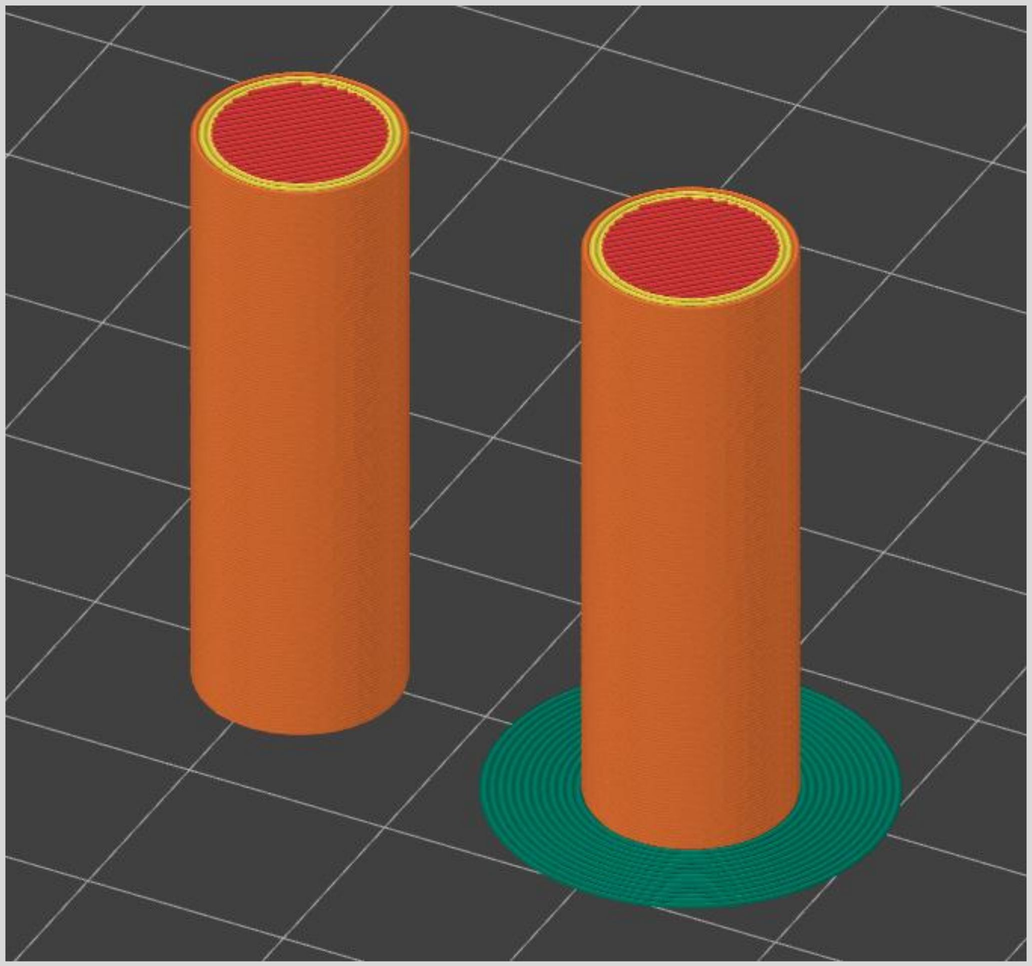
First Layer Warping

- Advice?
- **Better control over temperatures.**
  - Filament temperature.
  - Print chamber temperature.
  - Cooling fans.
- **Round corners to reduce stress concentration.**

(Picture from Creality Cloud Blog.)

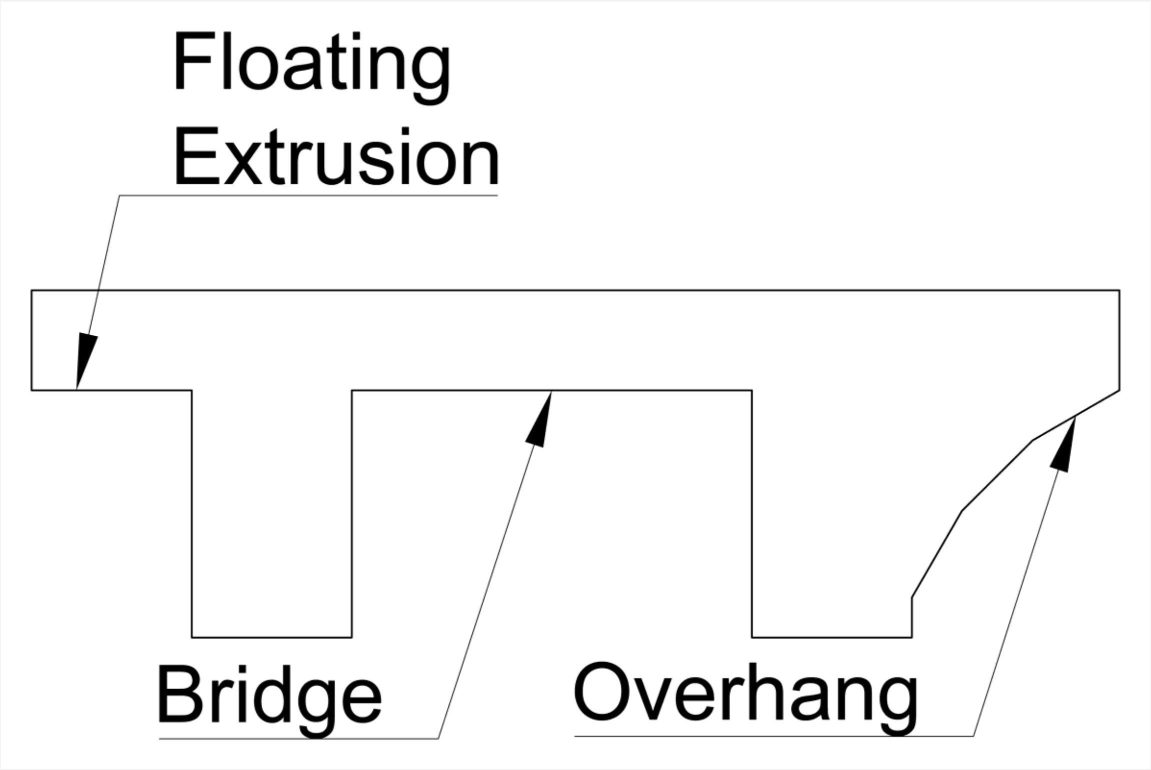
Brim

- Added during slicing.
- Improves first layer adhesion and fights against first layer warping.
- Concentric extrusions at the base perimeter. (Green lines on the right)
- One layer tall.
- Peeled or cut away after printing.



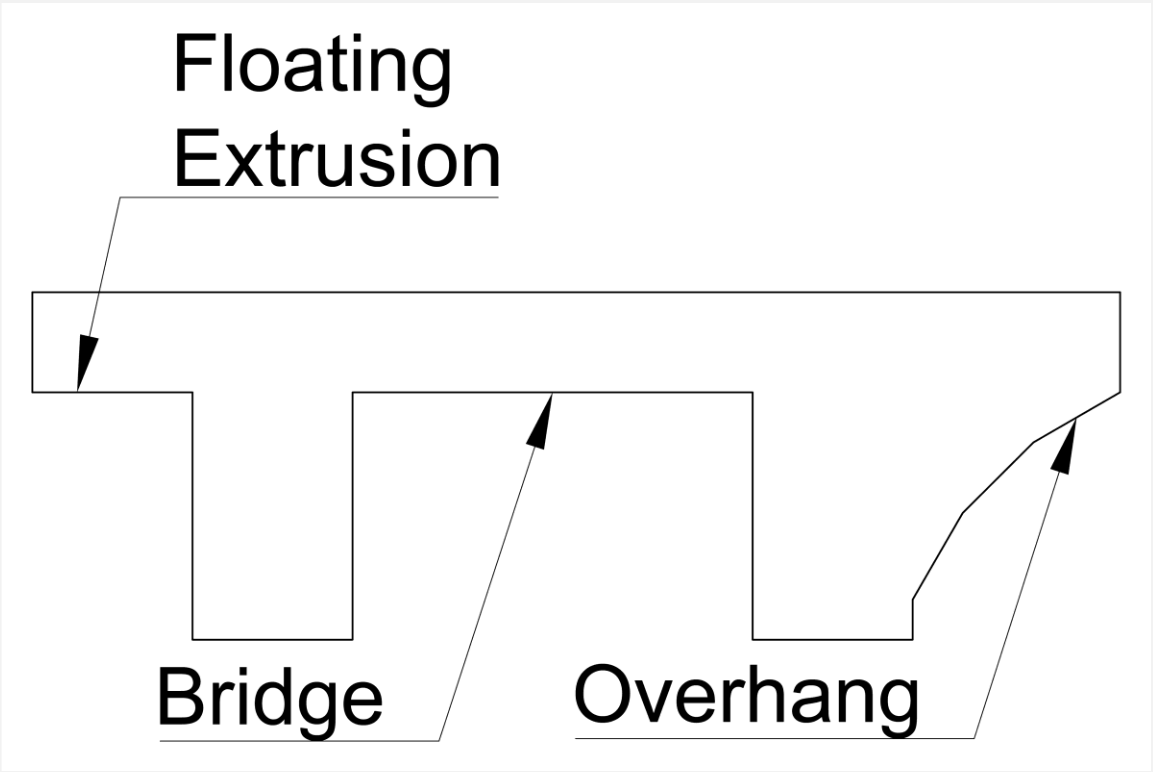
Fighting Gravity

- Why are prints limited by gravity?
- Extrusions do not levitate.
- **Extrusions work best when you press them onto a solid surface**, so some of this print is problematic.
- What parts will be the most deformed?



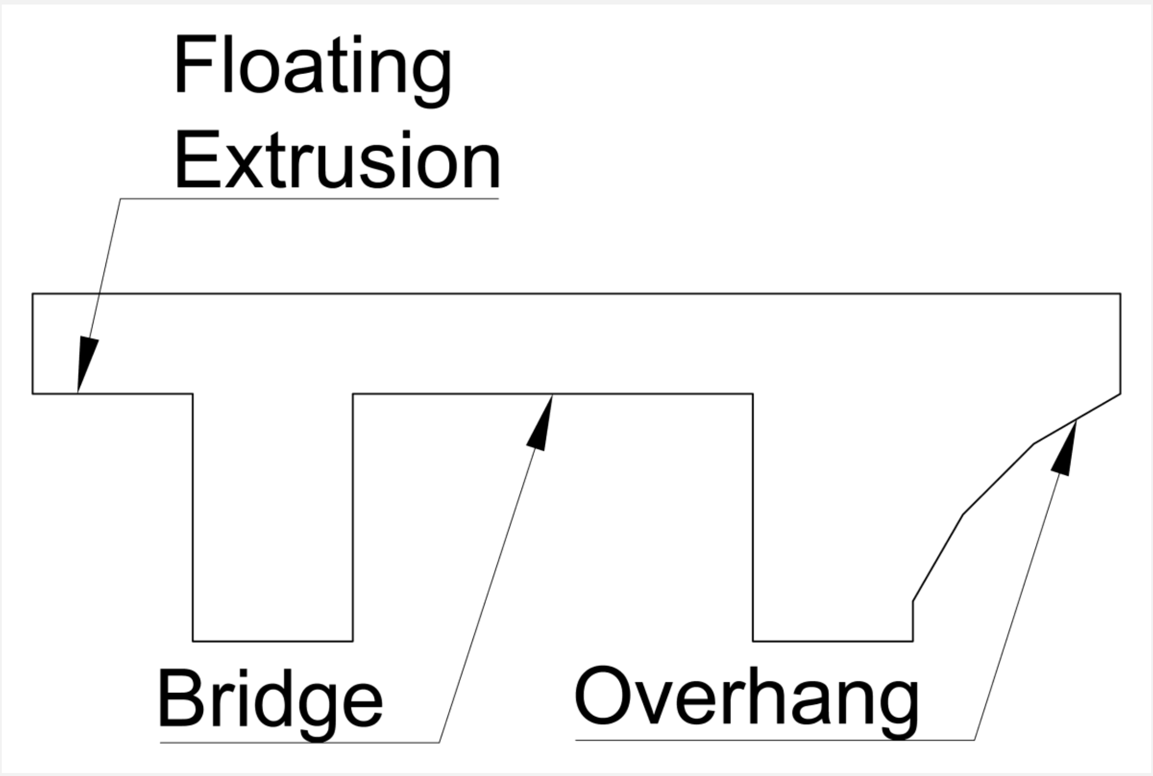
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Extrusion is supported along one side.
- Advice?
- **Keep overhangs at 45° to be safe.**



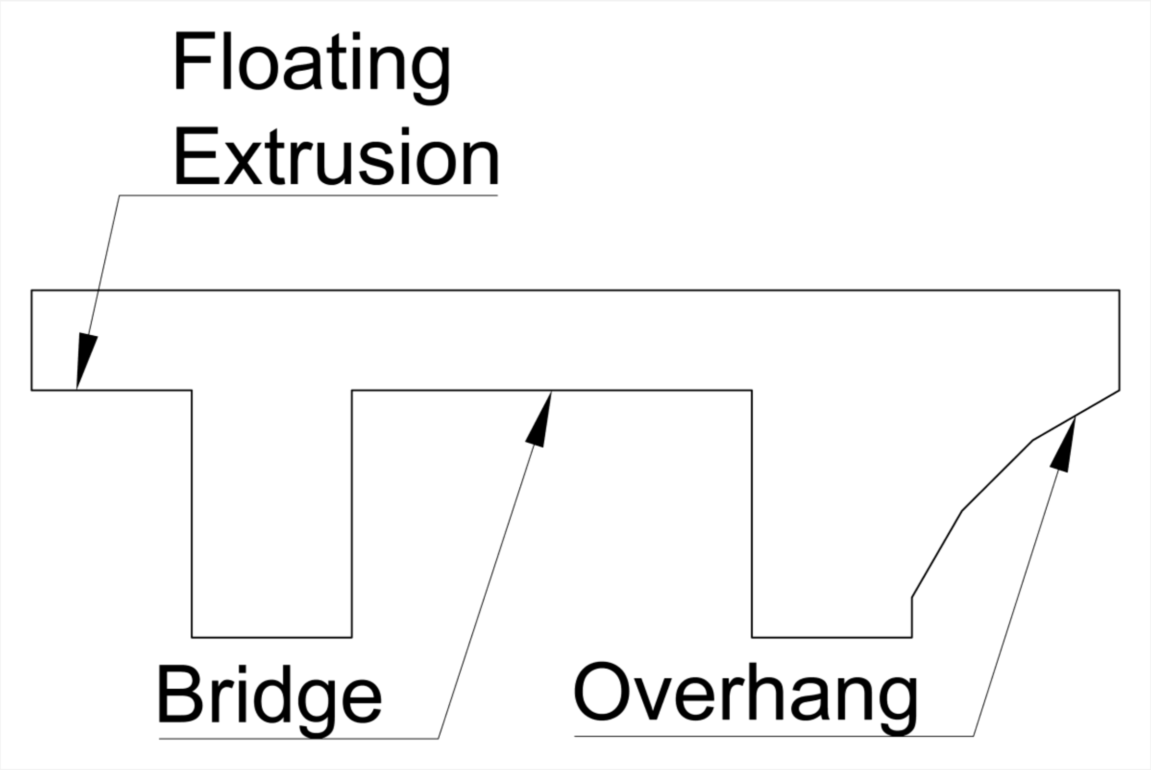
Bridge

- Both ends of the extrusion are supported and anchored.
- Middle hovers over a gap.
- The first few layers sag and droop down
- Extrusions eventually recover in most scenarios.
- Advice?
- **Keep bridges as short as possible.**



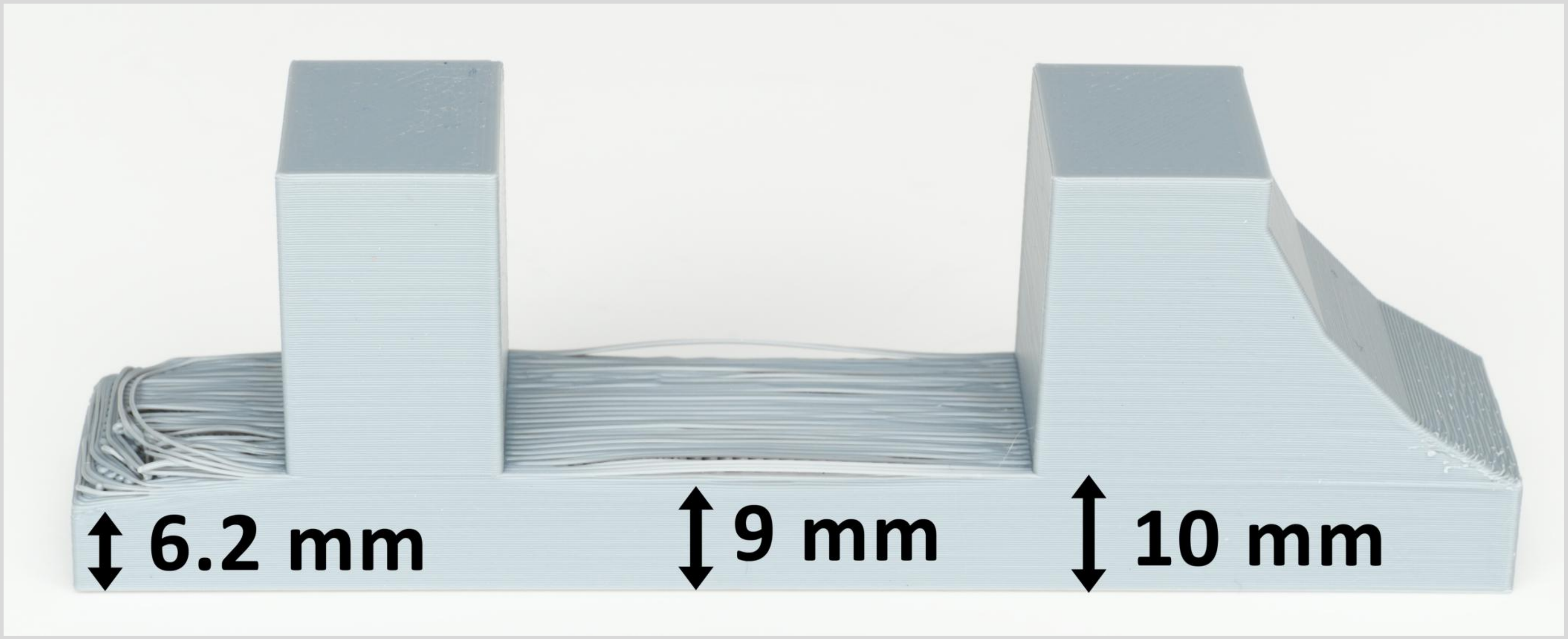
Floating Extrusion

- Extrusion that hangs in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid – very unreliable.**



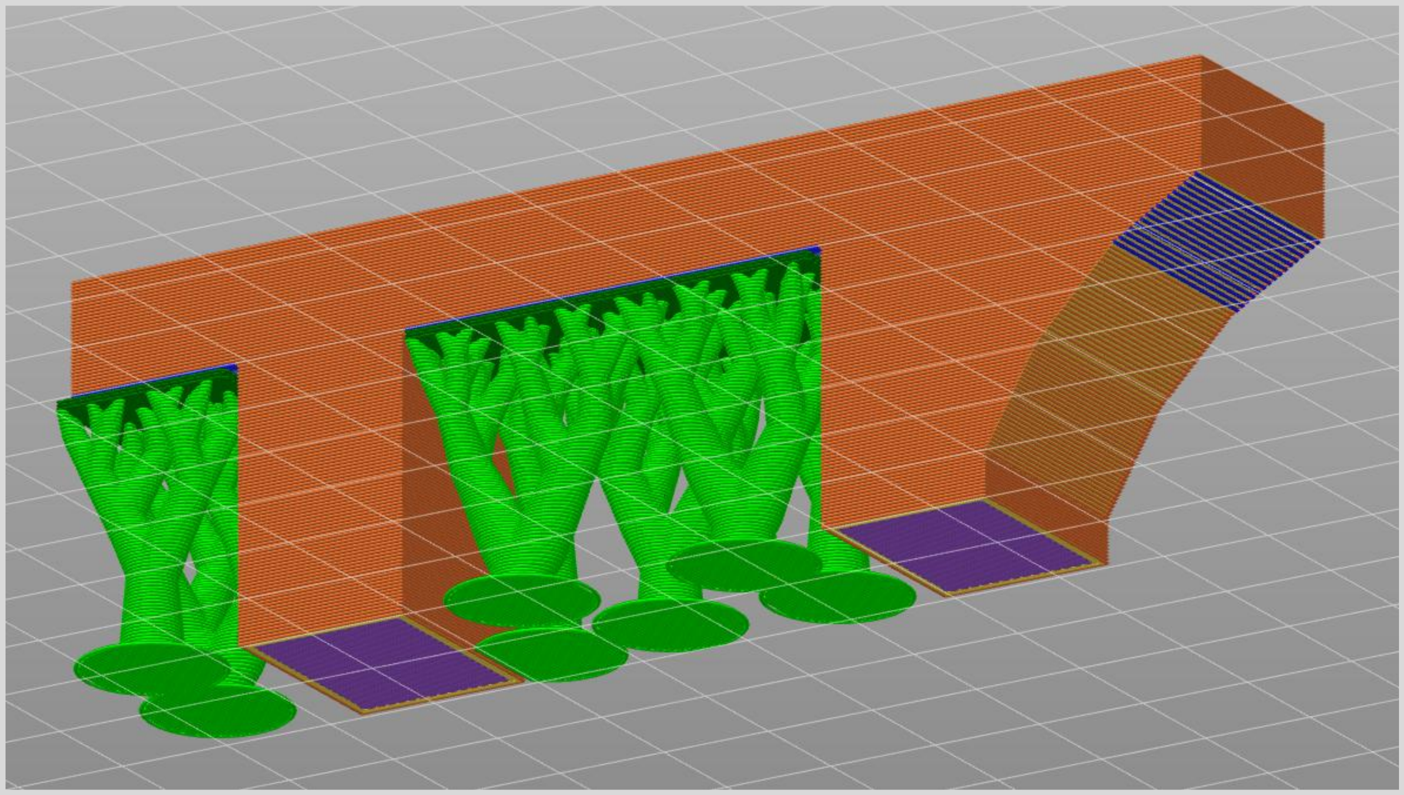
Test Print Results

- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Added by the slicer.
  - Green “trees”.
- Temporary scaffold that fixes:
  - Floating extrusions.
  - Long bridges.
  - Steep overhangs.
- Extra complexity, plastic, and print time.
- Tree/organic supports can lean and reach into spaces.

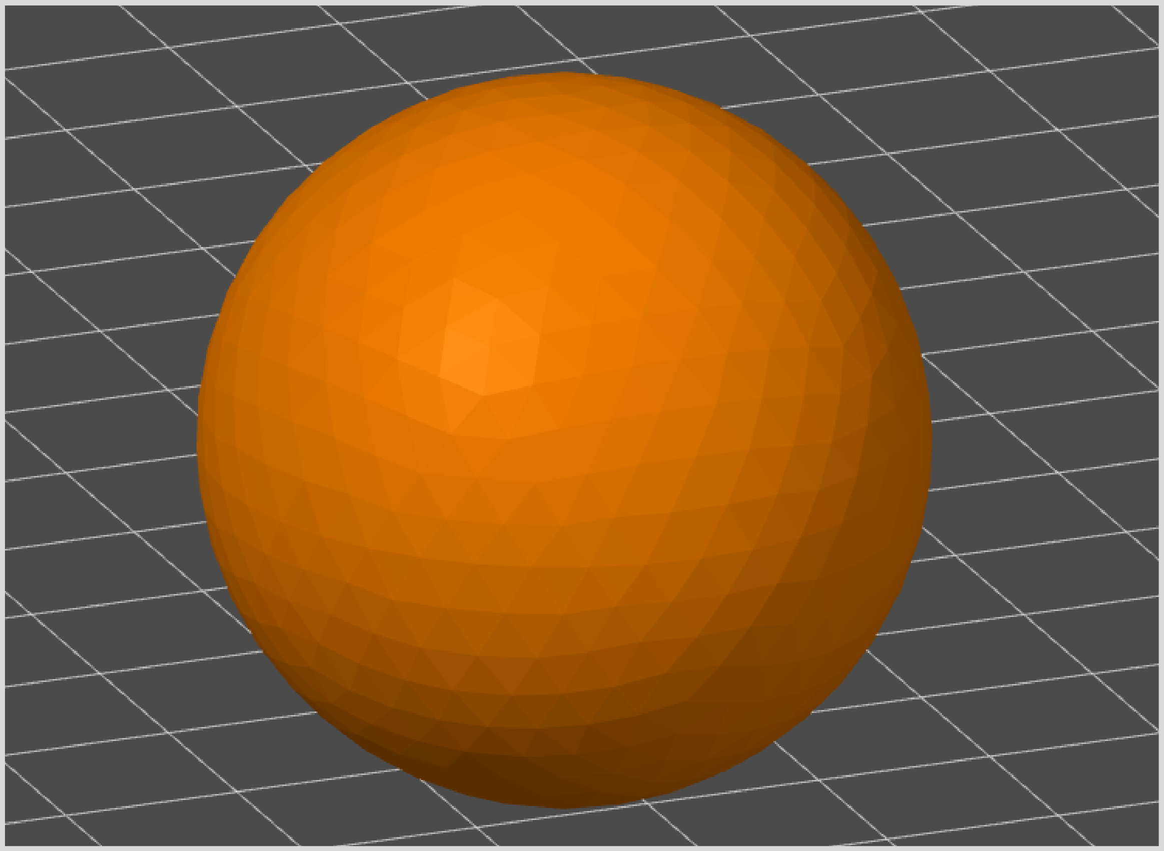


Supports

- In special situations, you can include supports as part of the 3D model.
- Supports are a delicate balance. Why?
  - If supports do not bond well enough … they hold nothing in place.
  - If supports bond too well … they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you have to cut away pieces of support.

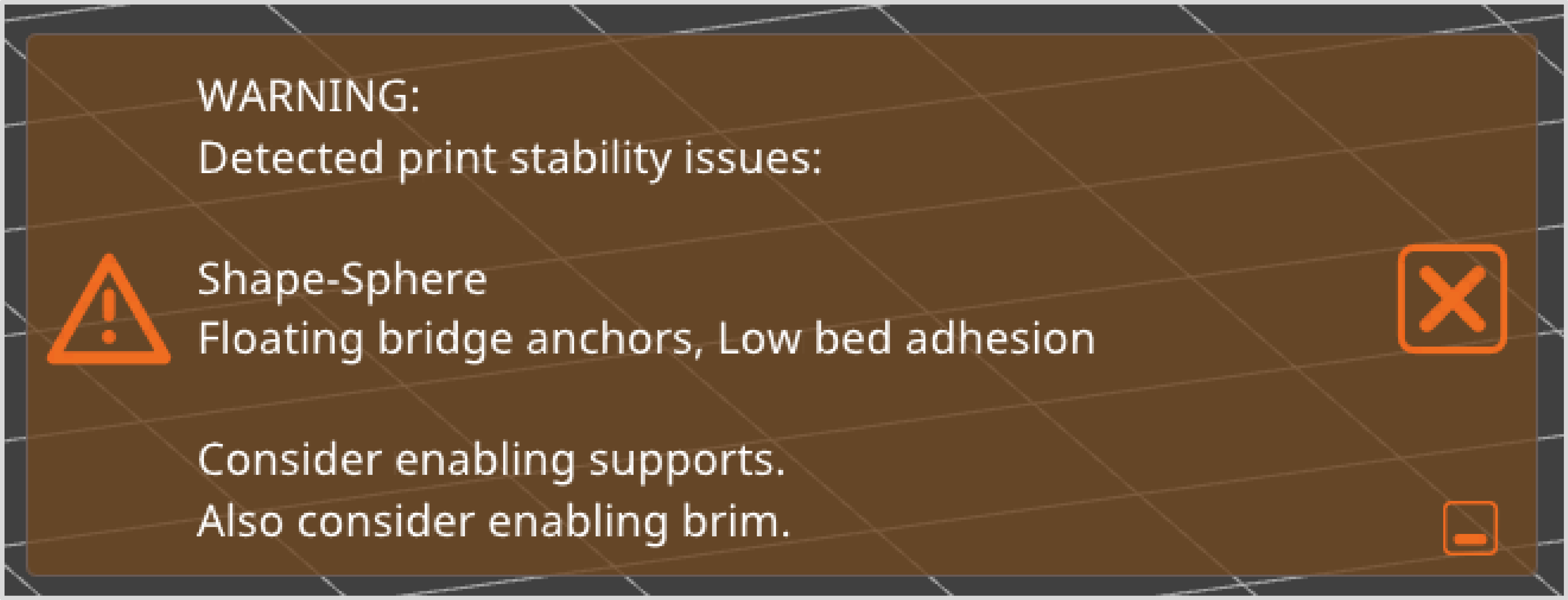
Limitations Review

- Printing a ball runs into some FFF limitations.
  - Very bad first layer contact.
  - Steep overhangs on the bottom.
  - The second layer is steep enough to be a floating extrusion.
  - Stair-stepping — uneven surface finish.



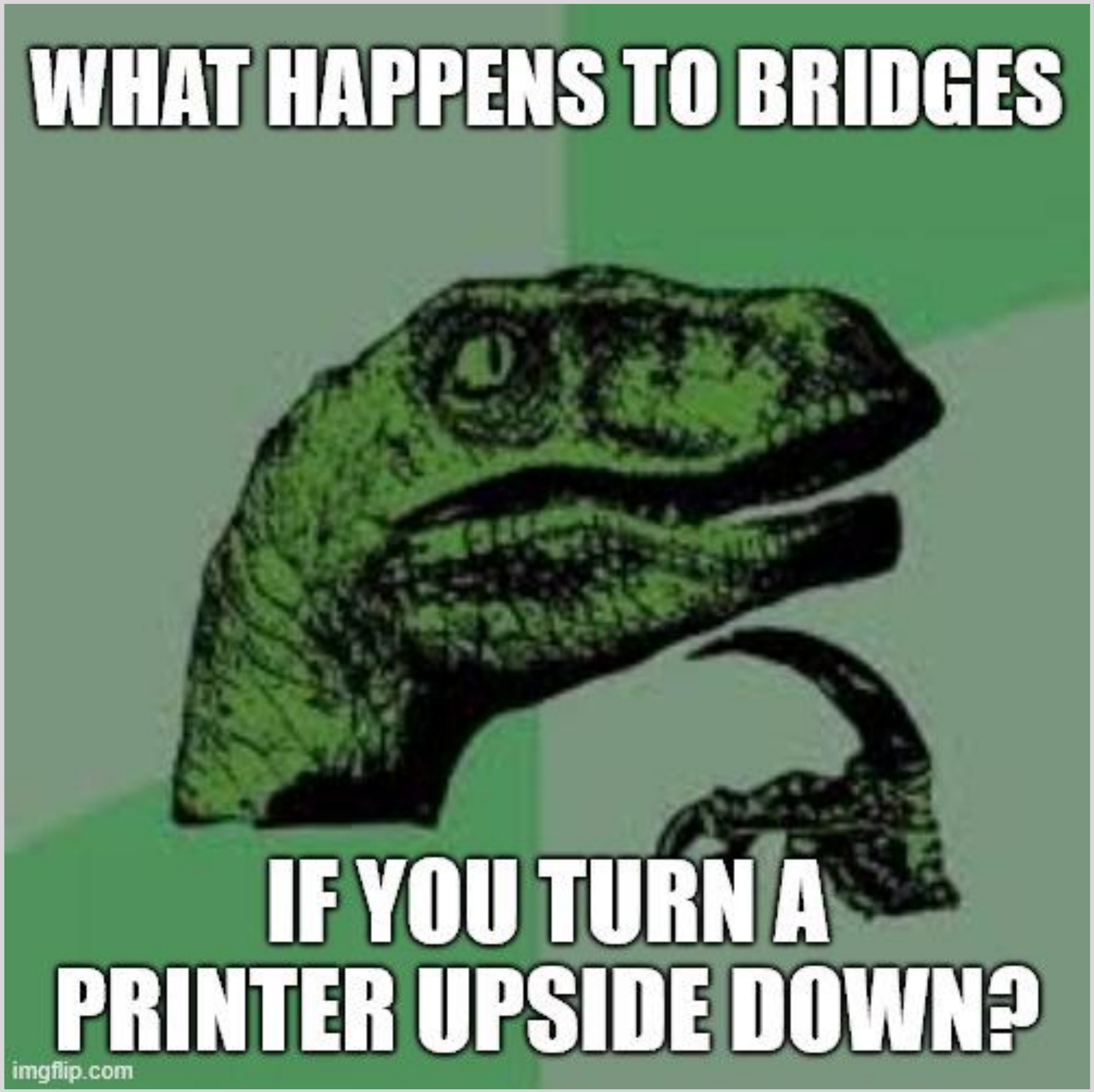
Limitations Review

- PrusaSlicer can recognize these problems.



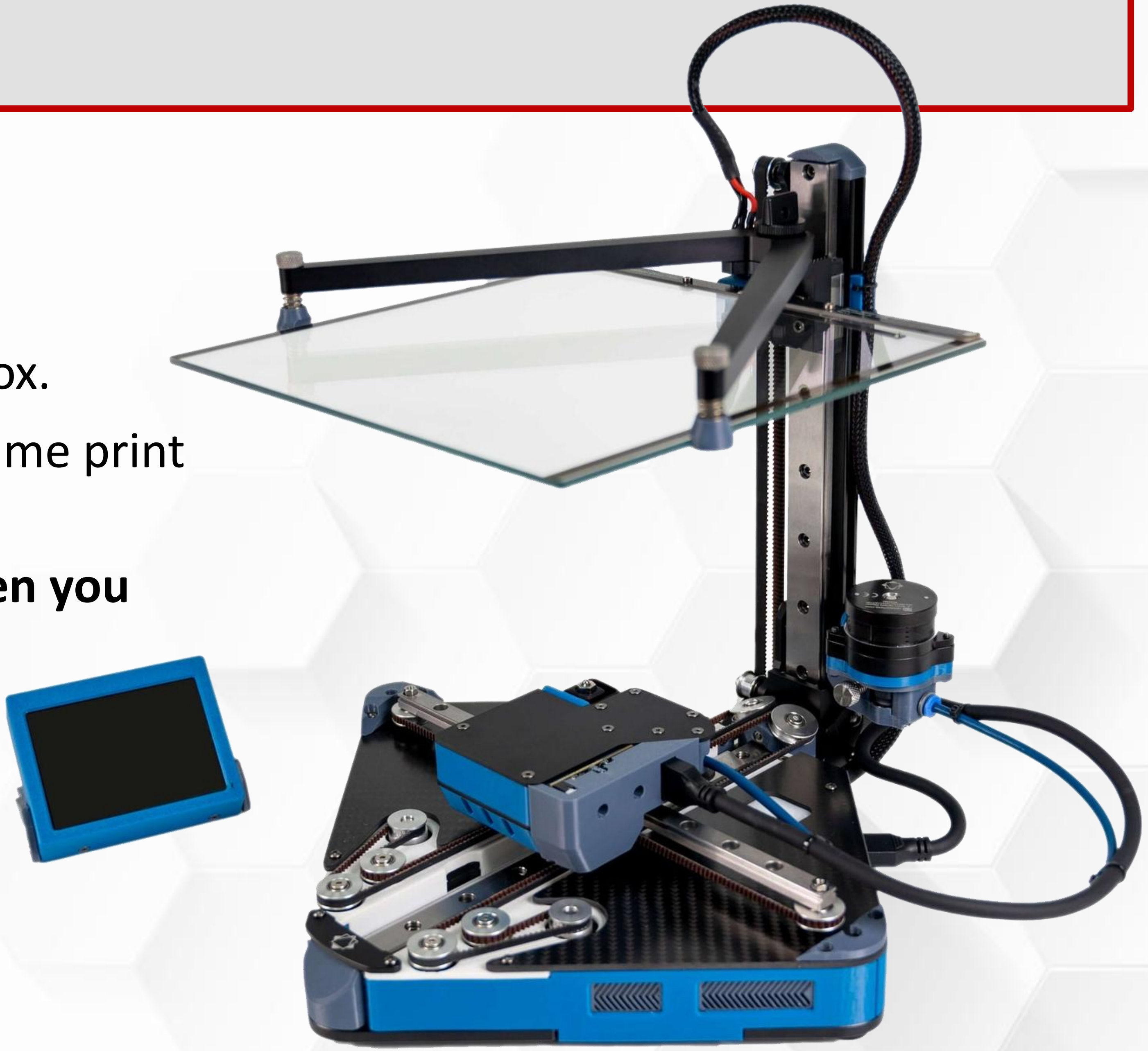
Limitations Review

- How can you fix this?
- Cannot fix stair-stepping by changing print orientation.
  - Use small layer heights to reduce visual artifacts.
  - Use adaptive layer heights to make the surface look more uniform.
- Brim is marginally helpful — the model starts with so little surface area.
- Add **supports** for most of the bottom.
- You could print two halves and then put them together.


WHAT HAPPENS TO BRIDGES
IF YOU TURN A
PRINTER UPSIDE DOWN?
imgflip.com

Anti-Gravity

- Positron Github.
    - Prints upside down.
    - Folds up and fits in a small box.
- Design team says: roughly the same print quality.
- **Extrusions hold their shape when you press them onto a solid surface.**



Dimensional Accuracy

- Be careful designing parts with exact dimensions.
- This is a common beginner mistake.
- Why?
  - Real life is filled with **error bars**!
  - Your prints are not exact.

Dimensional Accuracy

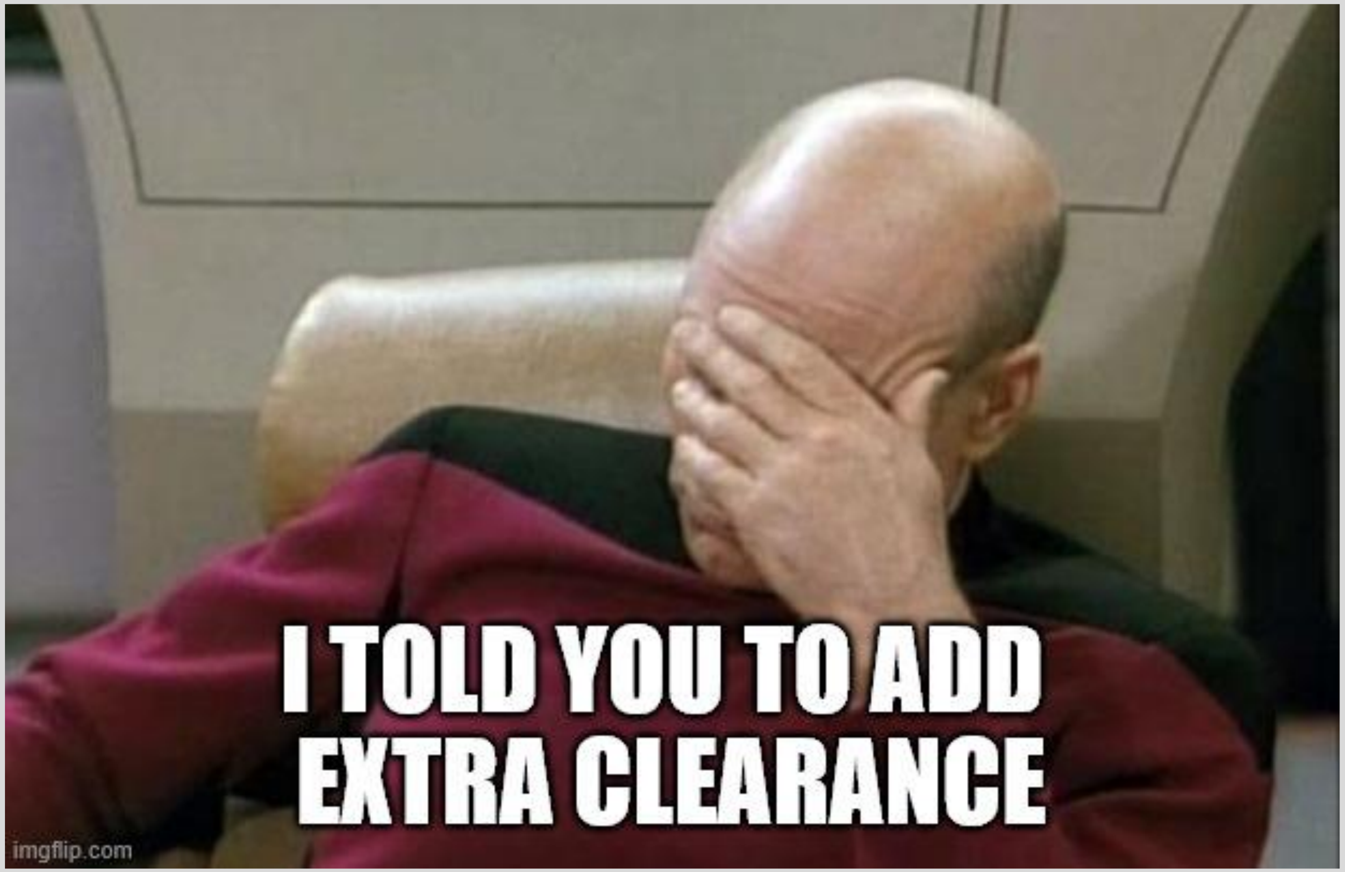
- How does this happen?
  - Shrinking and warping.
  - Improper hardware assembly.
  - Thermal expansion (discussed in Part 3).
  - Moving parts can wear out.
  - Loose screws.
  - Vibrations (discussed in Part 3).
  - Unoptimized slicer settings.

Dimensional Accuracy

- Advice?
- If you create a 3D design with multiple, assembled parts …
  - **Then you must add extra clearance (or air gap) between parts that touch.**
  - XY offset settings in slicer can give a similar (but inferior) result.
- How much clearance?
  - It depends, but not very much for modern printers.
  - *Example:* Making a hole for a 3 mm diameter bolt. Add 0.2 – 0.3 mm of clearance for a snug fit.
  - *Example:* Making a lid that easily slides into place. Add 0.4 - 0.8 mm of clearance for a loose fit.

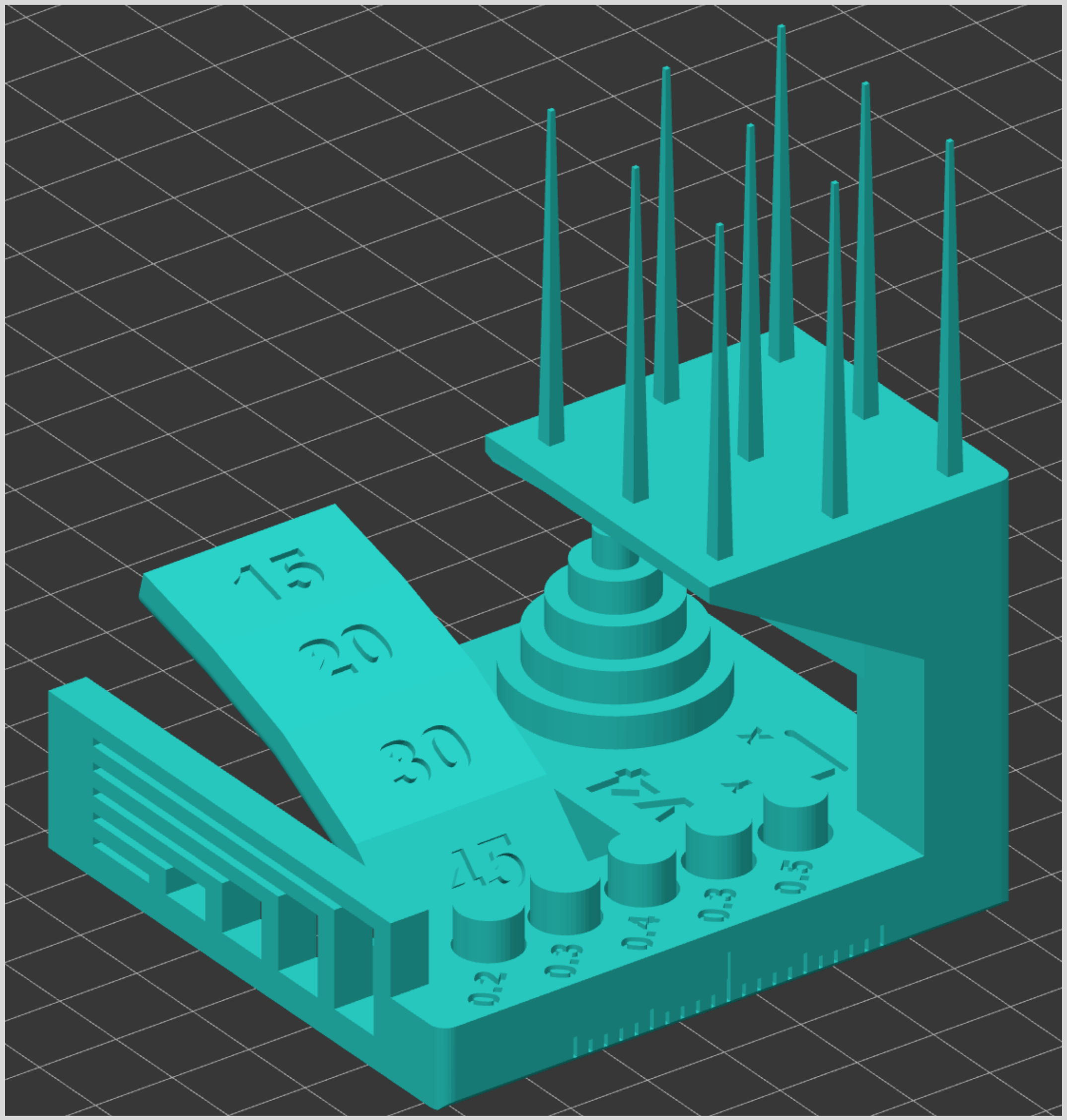
Clearance for Assembled Parts

- Do not forget!



Limitations Review

- Autodesk Test Print – One of many test prints.
  - What is each feature testing?
  - (GitHub link).
  - View and rate some completed prints in the “Makes & Comments” section.



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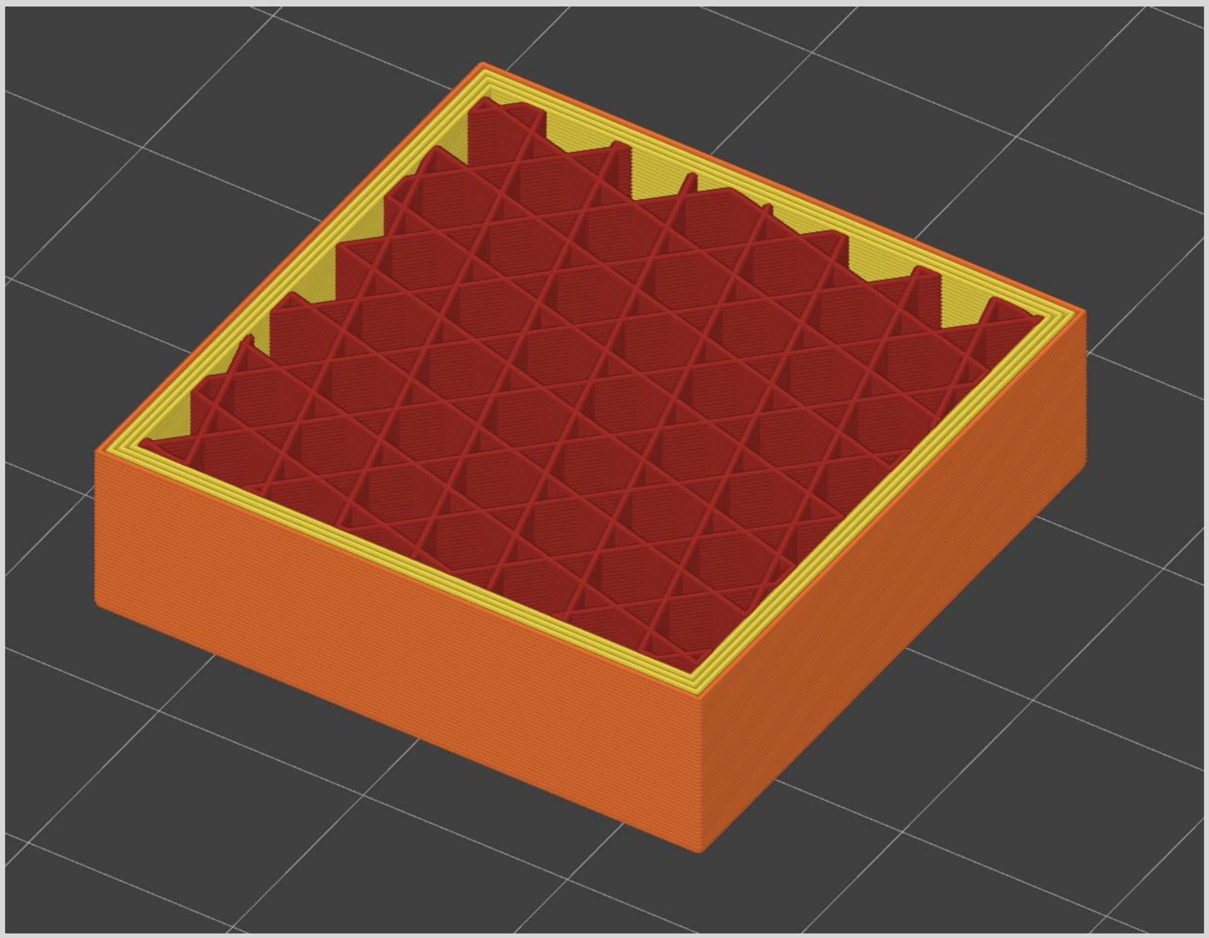
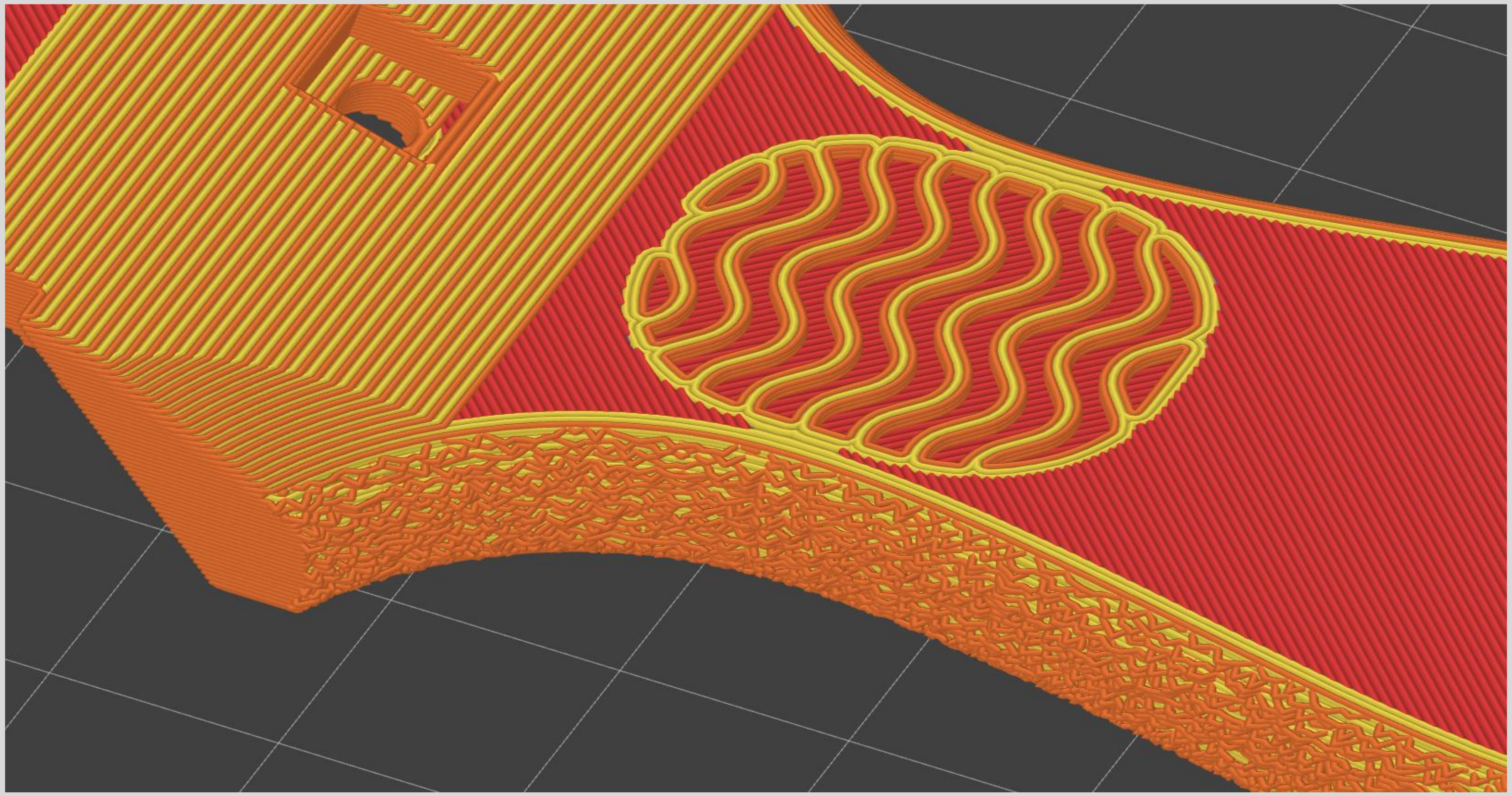
FFF Advantages

- Printers use tiny “building blocks.” Typical extrusions are
  - **0.2 mm tall** (1-2 pieces of paper).
  - **0.4 mm wide**.
- Not limited to uniform prints.
  - *Example:* Solid exterior and sparse infill.
  - Great at making hollow objects.
  - Imagine making this shape with wood or metal – much harder.



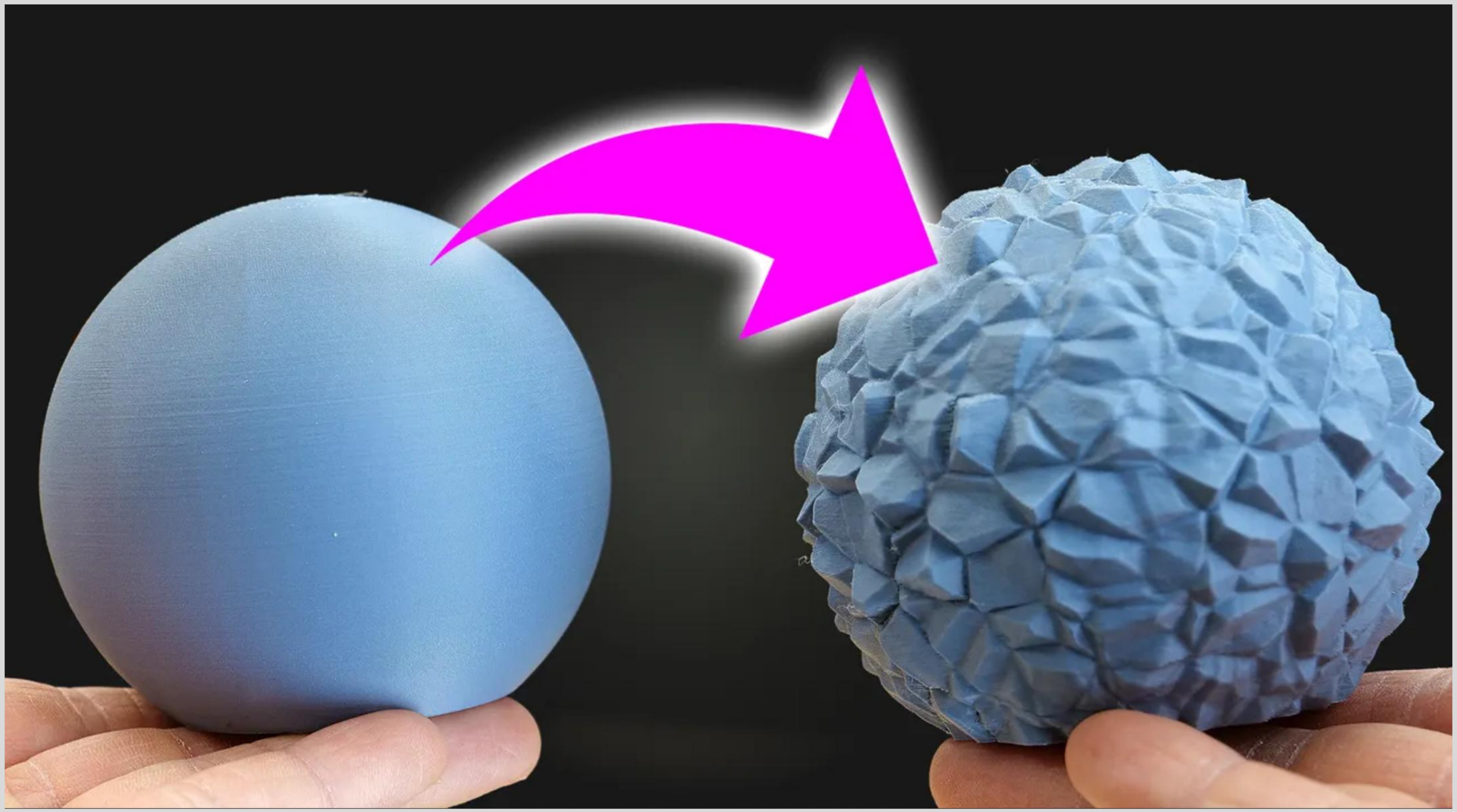
Custom Slicing

- Change slicer settings for specific geometry.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



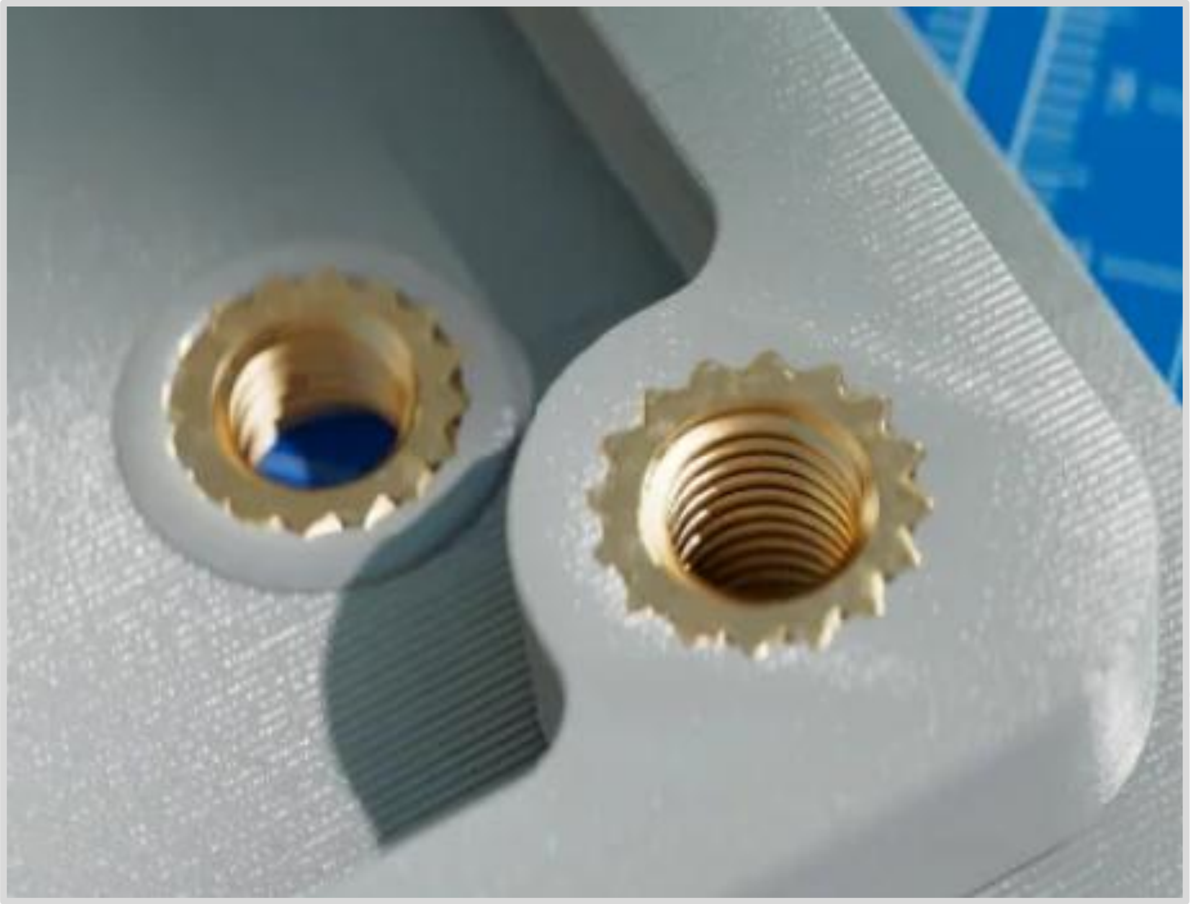
Community Projects

- BumpMesh is a free, open-source, browser-based tool for applying custom displacement textures to your 3D models.



Heat Set Inserts

- Plastic is soft → Screws wear out plastic holes quickly.
- Threaded anchors for screws and bolts.
- Installed with a hot soldering iron.
- Very durable.



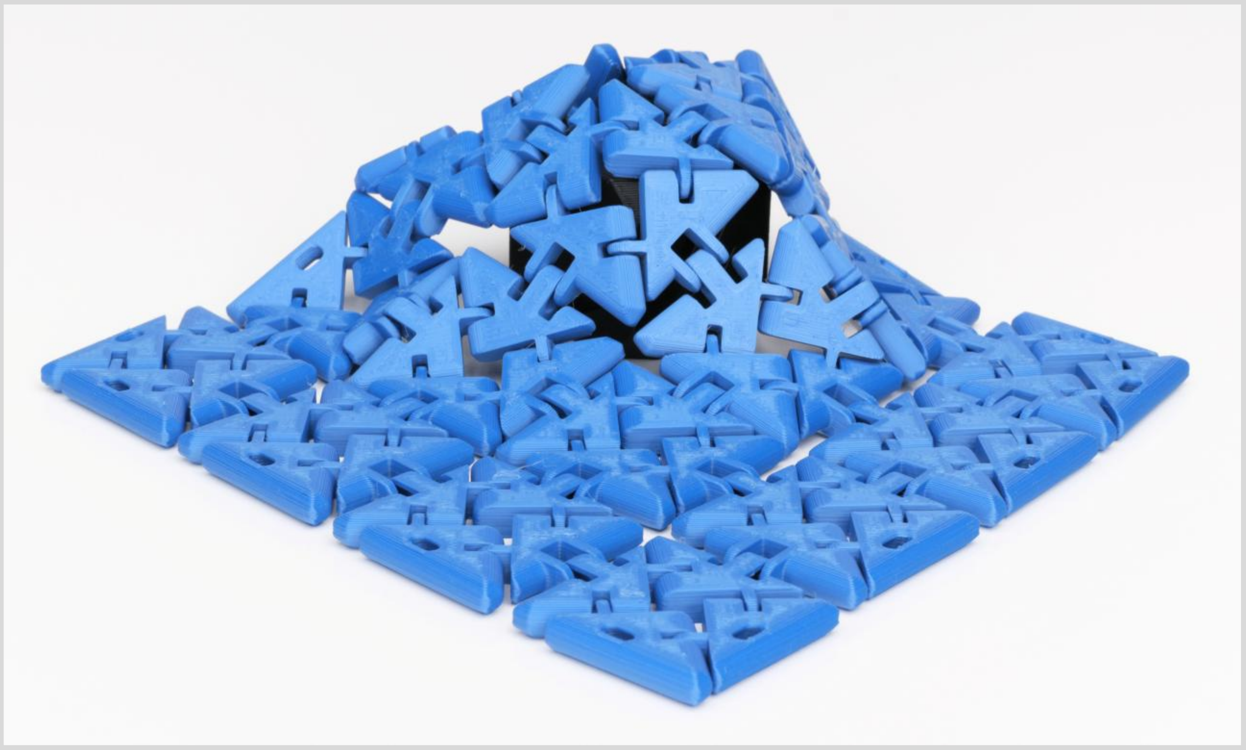
(Pictures from CNC Kitchen.)

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
  - Nuts and fasteners.
  - Magnets.
  - Weights.
  - Reinforcing plate.
  - Carbon fiber rods.
- Object is locked inside when the print finished.
- Add a pause command in the gcode.
- No part of the object can sit above the current layer, or it will create a nozzle collision.
- Waiting around for the right layer is inconvenient.

Print-in-Place (PIP)

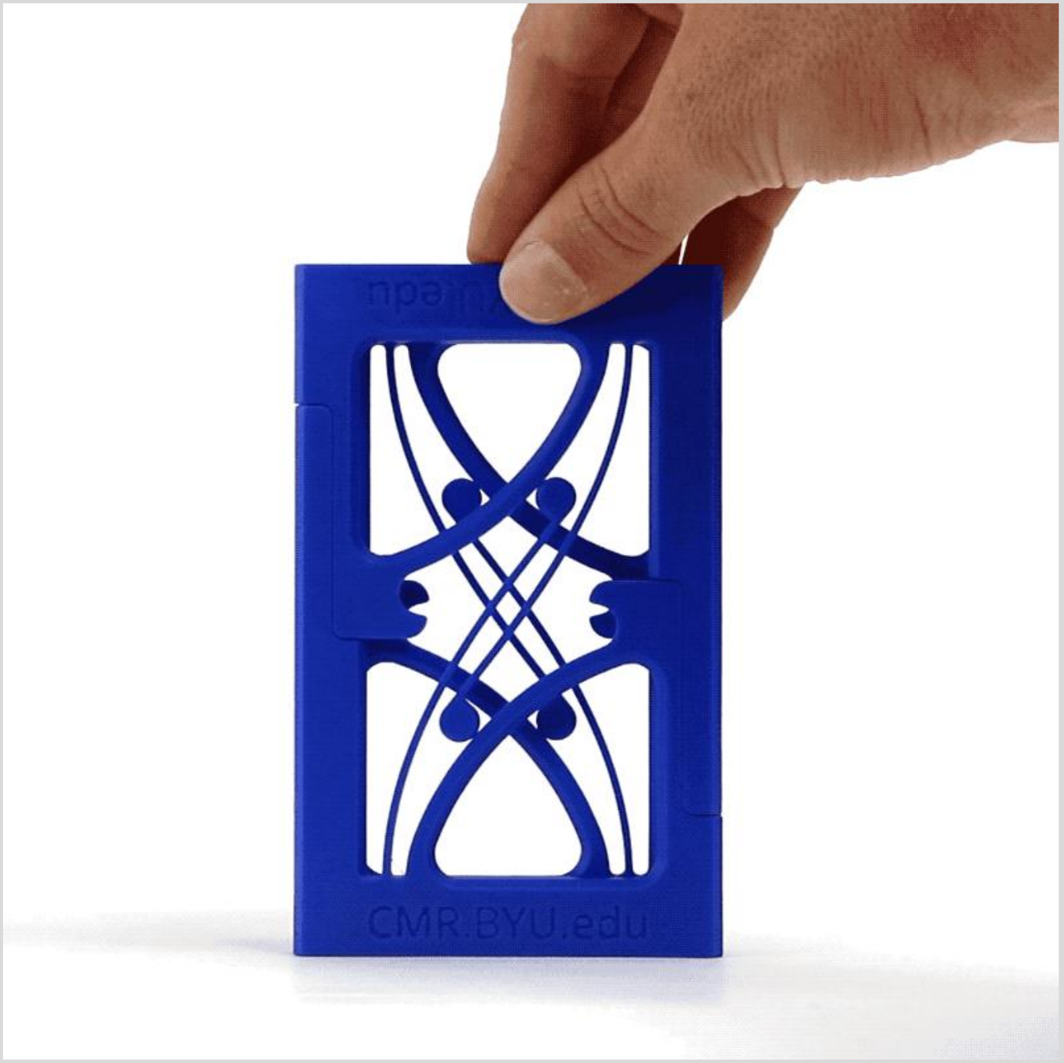
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by printables.com/@FlyingGyroscope



Functional Flexing

- Springs and **compliant mechanisms**.
  - Features/shapes that function by bending and flexing.
  - Elastic materials (springy, without permanent deformation).
- Squeezing springs is literally stressful.
  - Pay attention to layer lines for strong parts.
  - For example, these parts were printed laying flat (not standing up).

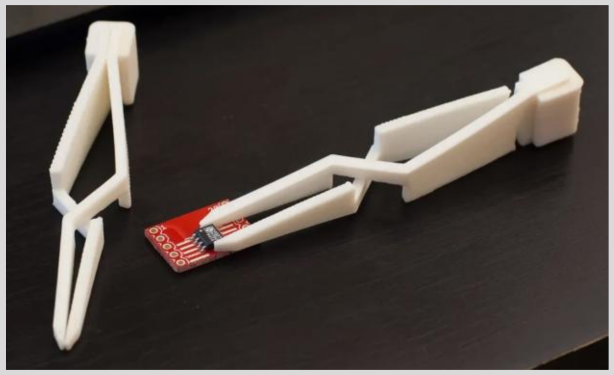
(Constant Force Mechanism by BYU CMR.)



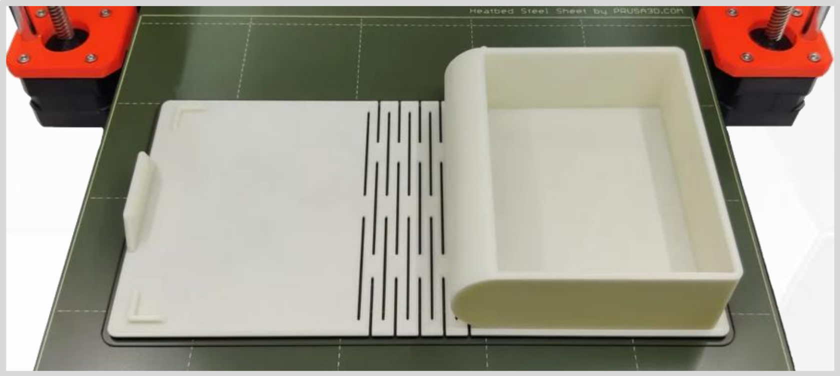
Springs by VC Design

Clip by AE

Cross tweezer by joo



Living hinge by DrJones




THANK YOU
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